Replication Paper

Liberals Lecture, Conservatives Communicate: Analyzing Complexity and Ideology in 381,609 political speeches

Schoonvelde, Brosius, Schumacher & Bakker (2019)

Sara Murphy, Text Analysis, March 30, 2026

Introduction

- **Impact of political party ideology on linguistic complexity**
  - **Hypothesis:** Liberals are more "open minded" and talk with more tolerance for ambiguity; Conservatives are more "conscientious" and desire closure, talking more short and direct

- **Key Findings:**
  - More "culturally liberal" political parties use more complex language
  - Mostly no difference in linguistic complexity due to "left/right" economic political differences
  - Mixed impact of being the "governing party" on linguistic complexity – varies by country, context
  - Downward time trend in complexity (complexity decreasing over the years)

- **Limitations:**
  - Within country comparisons only – not between countries
  - Complexity score (Flesch Kincaid Grade) most attuned to English language
  - Scale versus human validation

Data: Sample

- **Research Goal:**
  - Replicate beyond US/UK
  - Account for contextual factors (audience of speech, status of party, time/decade)

- **Sample:**
  - 381,609 speeches, Europe
    - 10 European countries
    - 70 years (1945-2015)
    - 12 European prime ministers
  - Speeches with 11+ sentences with 5+ words

- **Datasets (3), Corpora (8)**
  - **Parliamentary Speeches** x 5 Parliaments
    - Audience: Other MPs
      - Britain (161,683, 1988– 2015)
      - Germany (66,061, 1991-2013)
      - Holland (48,546, 1994-2015)
      - Spain (35,986, 1989-2015)
      - Sweden (72,999, 1991-2015)
  - **EU Speeches**
    - Audience: Politicians, foreign officials, citizens
      - 1,847 speeches, 10 countries
      - 2007-2015
      - English language speeches only
  - **Party Congresses** X 2 countries
    - Audience: Internal Party Members
      - Denmark (528 speeches, 5 political parties)
      - Netherlands (659 speeches, 7 political parties)
      - 1945-2017

Data: Variables

- **Unit of analysis**
  - Individual speech

- **Dependent Variable: Linguistic Complexity**
  - Words, sentences and syllables
  - **Use Flesch-Kincaid Grade score (FK)**
    - Higher FK = higher complexity

- **8 OLS regressions**
  - 5 for parliamentary corpora (5 countries)
  - 2 congress speech corpora (Denmark and NLD)
  - 1 heads of government (EU)

Flesch–Kincaid Grade Level

$$0.39\left(\frac{\text{total words}}{\text{total sentences}}\right)+11.8\left(\frac{\text{total syllables}}{\text{total words}}\right)-15.59$$

Data: Variables

- **Independent Variables:**
  - **Left-right ideology (economic)**
  - **Liberal-conservative ideology (social)**
  - Time (year)
  - Speaker = government party (1,0)

- **Constructing Ideology: Manifesto Project Database**
  - 53 issues, sentence level
  - Last year's election manifesto (t -1) = party = speaker ideology (t)

- **Authors created two left/right splits from Manifesto Project**
  - **PC** = conservative issues (anti-EU, anti-immigration, pronational way of life, pro-traditional morality, anti-multiculturalism, pro-military, anti-internationalism, pro-Freedom and Human Rights and pro-political authority, pro-law and order); liberal issues (pro-EU, pro immigration, anti-national way of life, anti-traditional morality, pro-multiculturalism, antimilitary, pro-internationalism, anti-imperialism, pro-peace, pro-environment, pro-culture, and support for under-privileged minority groups).
  - **LR** = Left-wing = market regulation, economic planning, corporatism, protectionism: positive, Keynesian demand management, controlled economy, nationalisation, marxist analysis, welfare state expansion, education expansion and support for labour groups. Right-wing = free-market economy, incentives, protectionism.

Replication Dataset

- Data & Replicable Code: Harvard's Dataverse

- **3 Phases:**
  - **1) Replicate from Pre-Processed Data**
    - Full code
    - Full dataset
    - No raw speech data

  - **2a) Replicate Linguistic Complexity** (Flesch Grade score) - **Parliament**
    - 4/5 Corpora available; sample portion of text ~ 4% of speeches overall
    - Raw speech data

  - **2b) Replicate Linguistic Complexity** (Flesch Grade score) **– EU and Congress**
    - 1/1 Corpora for EU
    - 1/2 Corpora for Congress (Netherlands Only)
    - Raw speech data

Replication: Phase 1

Pre-Processed Data: All Tables and Figures

Figure 1 – Mean Complexity by Select Speaker (Parliament)

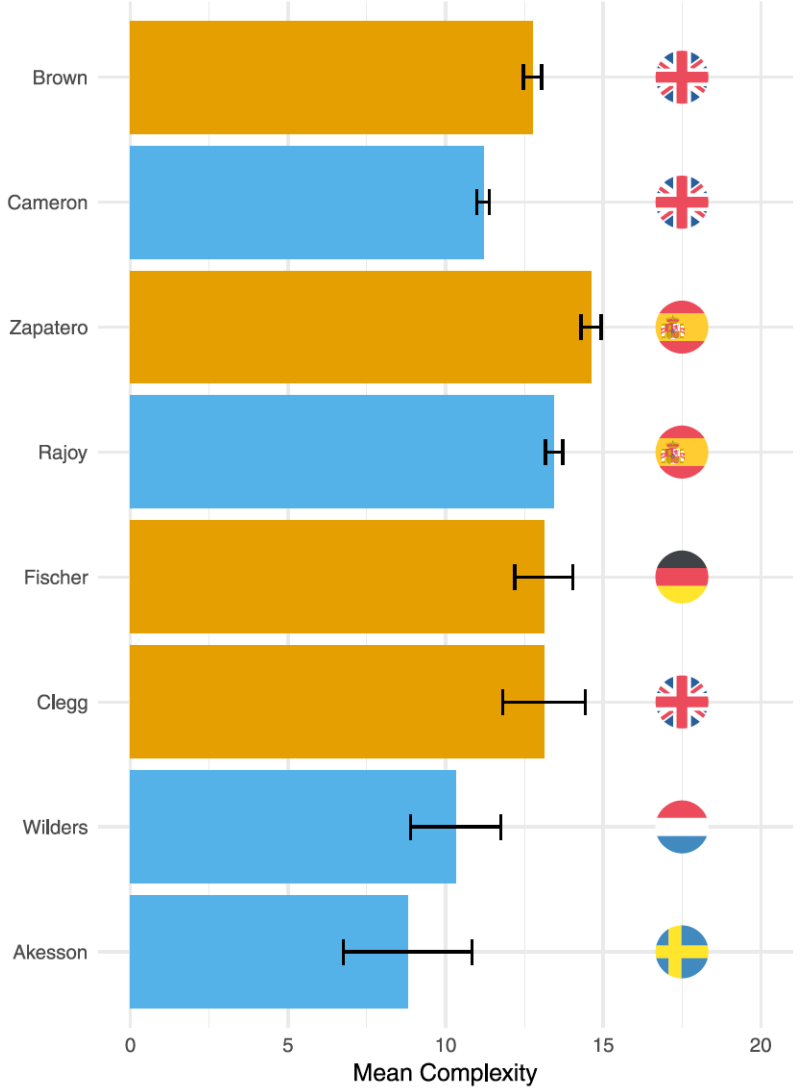
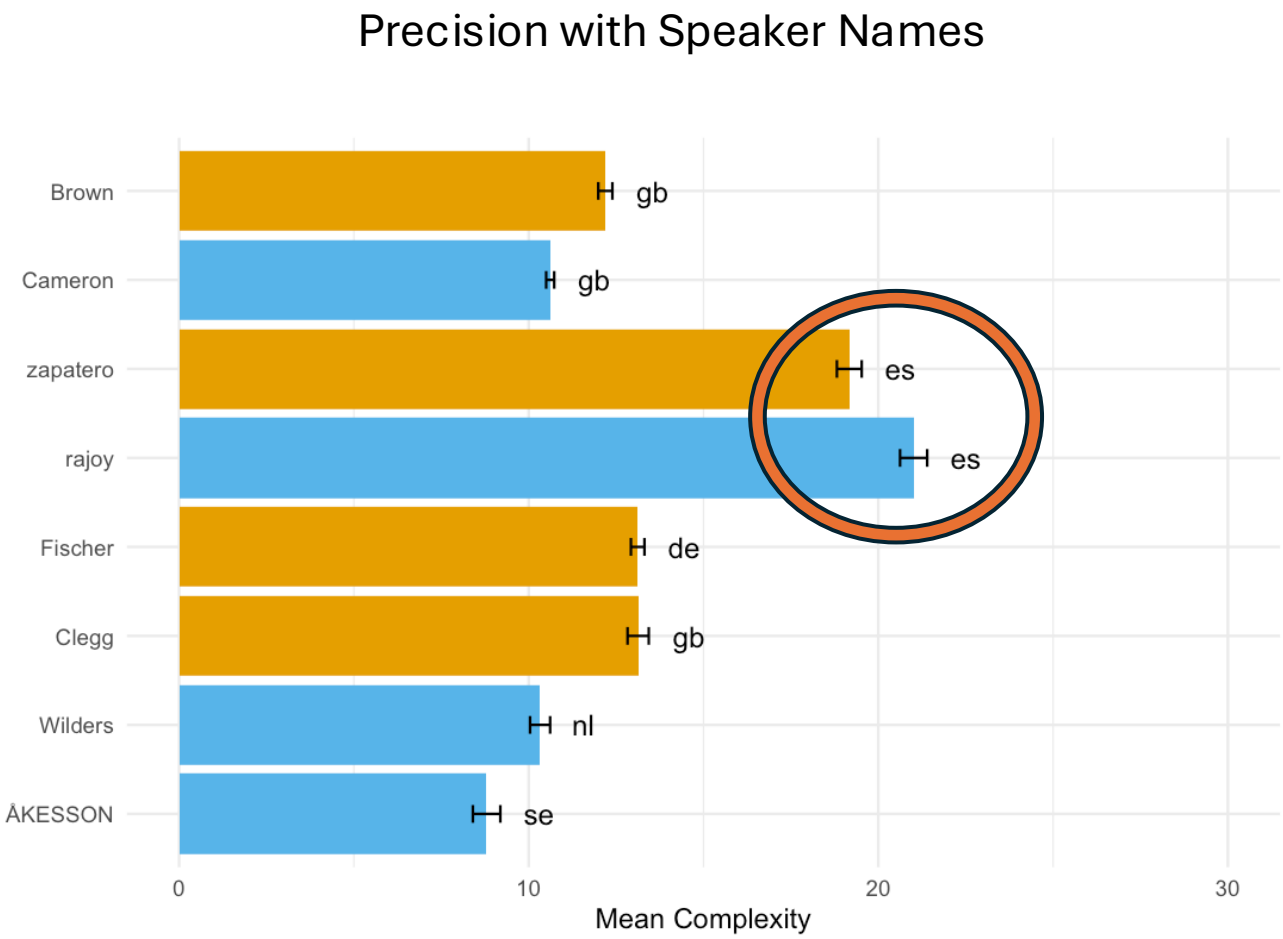
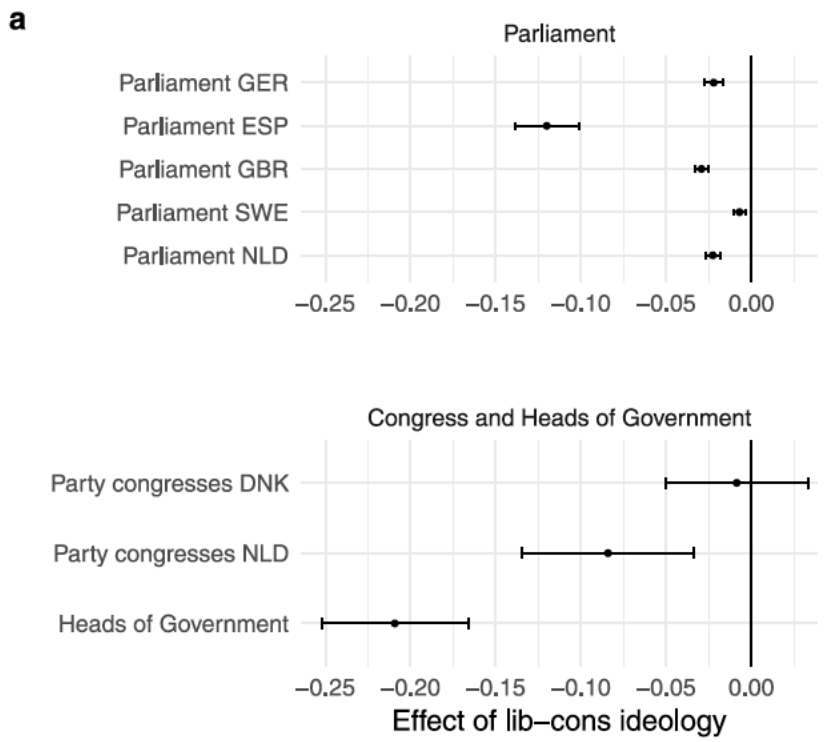


Fig 1. Descriptive information on linguistic complexity. The bars in this figure denote mean complexity scores, with 95% confidence intervals, of selected speakers.



Replication

Figure 2a – OLS Regression: Complexity on Ideology (Libs-Cons Social Ideology)



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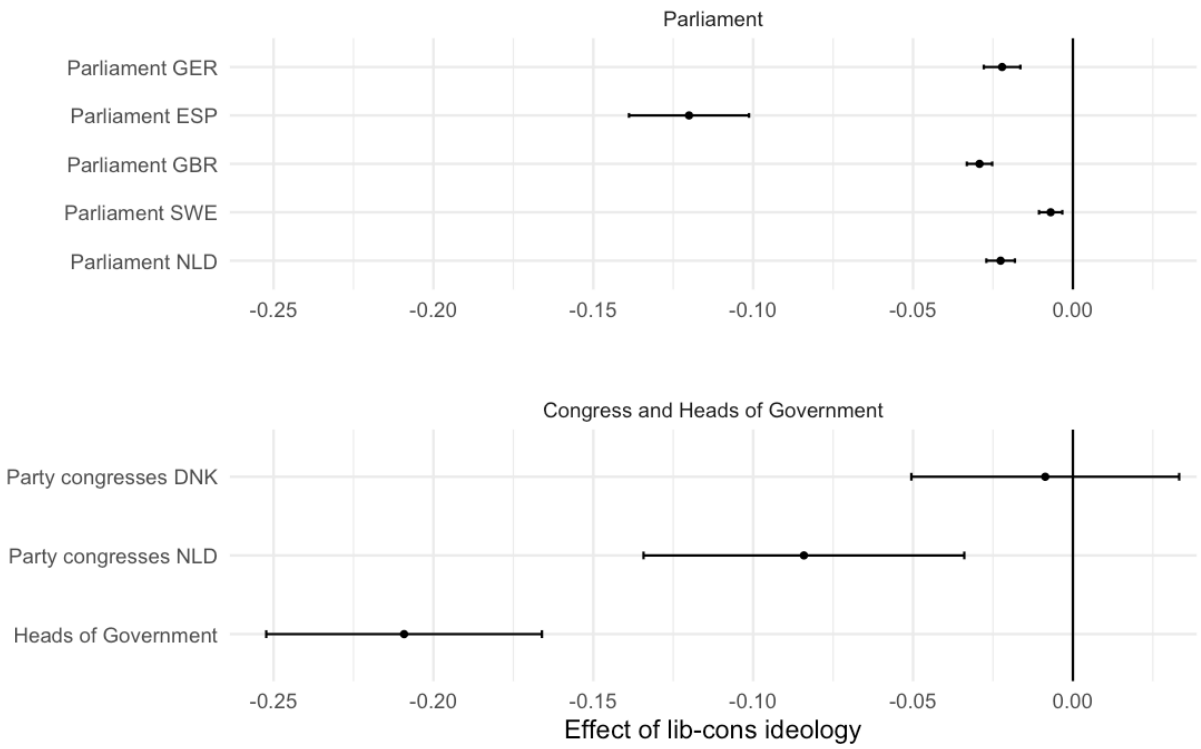
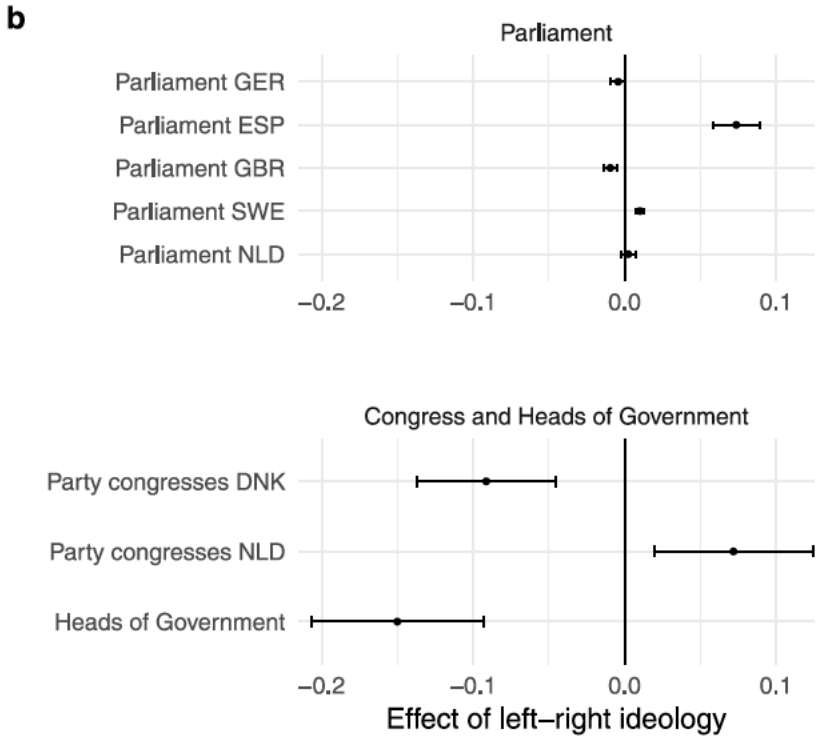


Replication

Figure 2b – OLS Regression: Complexity on Ideology (Left-Right Economic Ideology)



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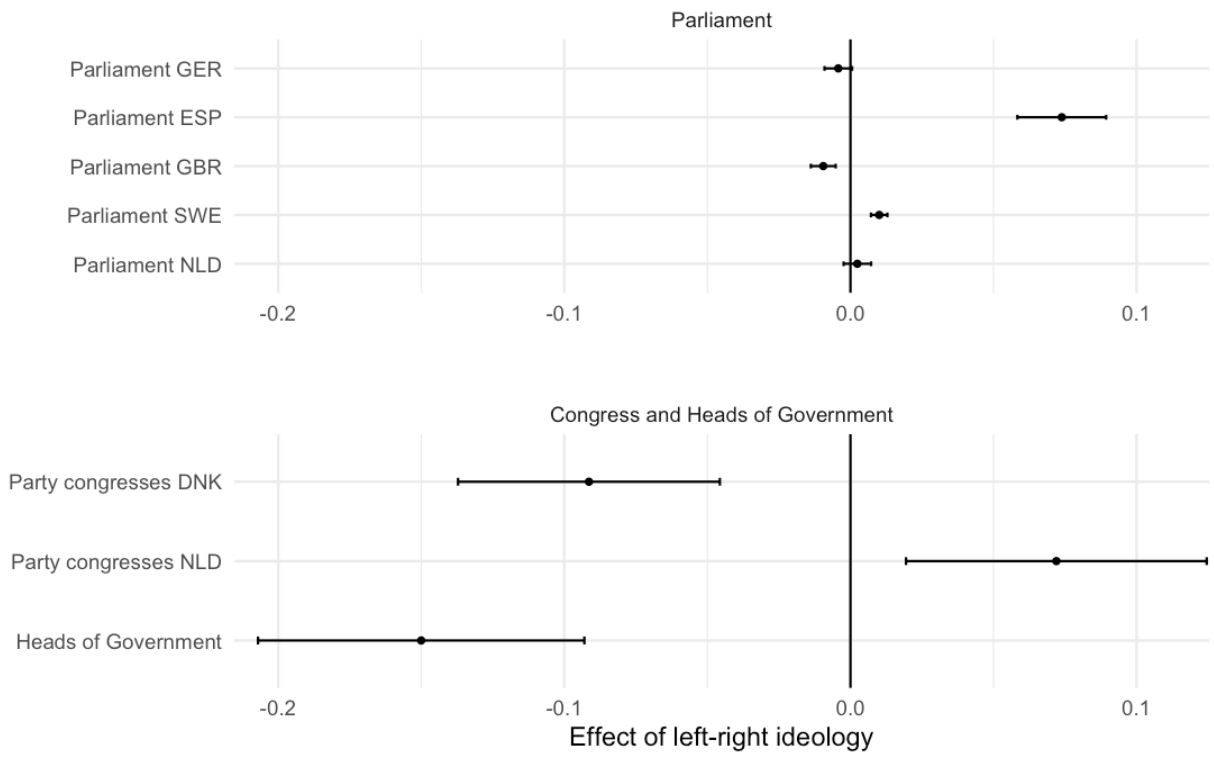
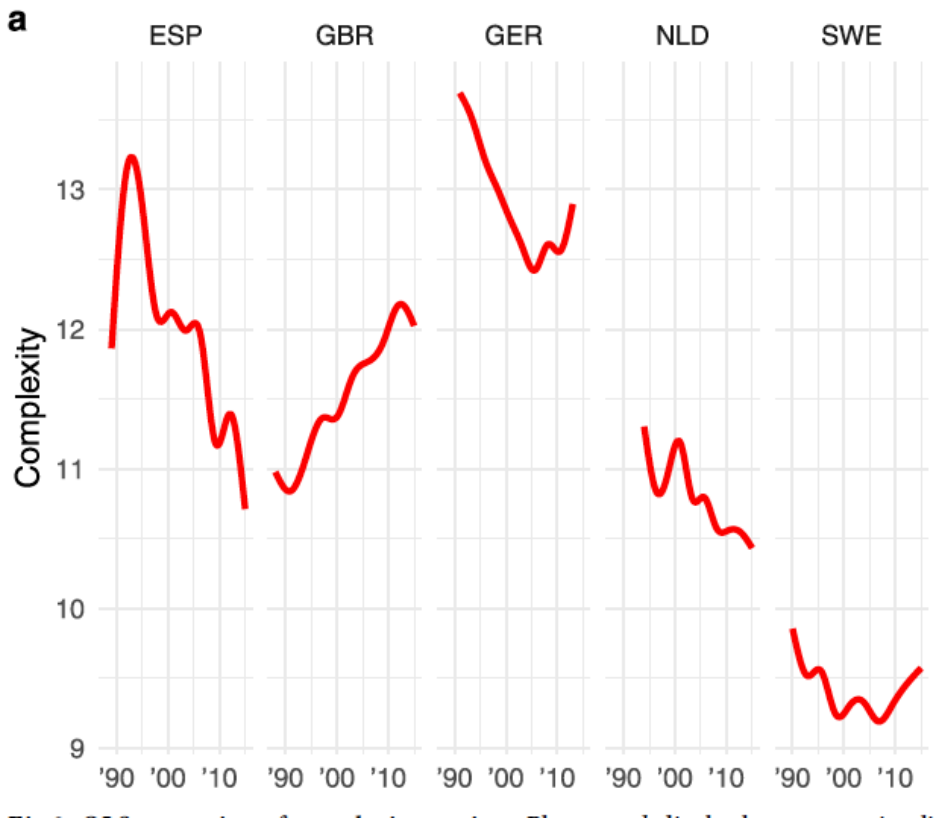


Replication

Figure 3 – OLS Regression: Complexity on Time



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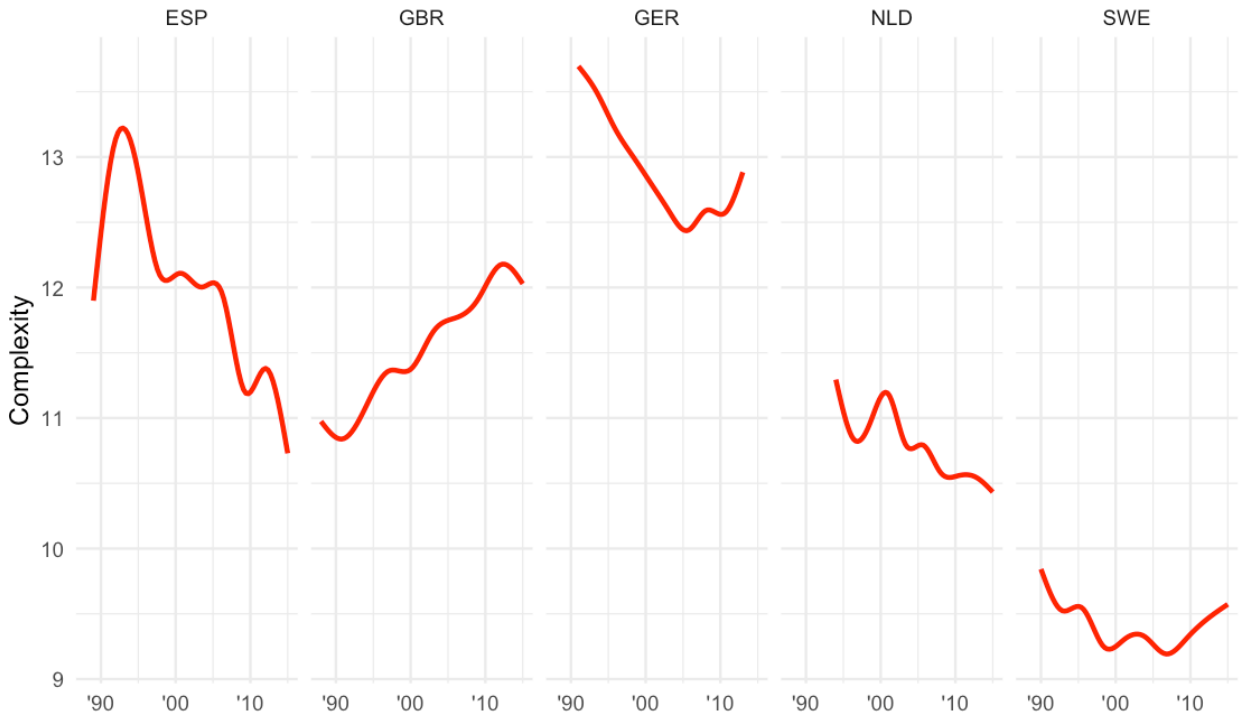

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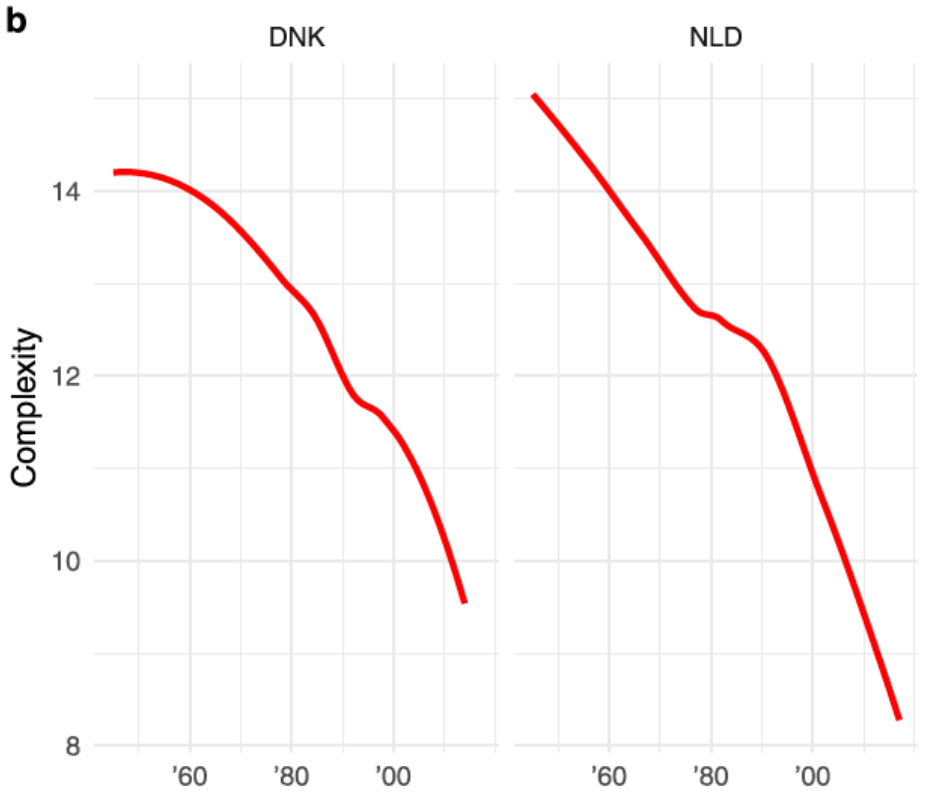

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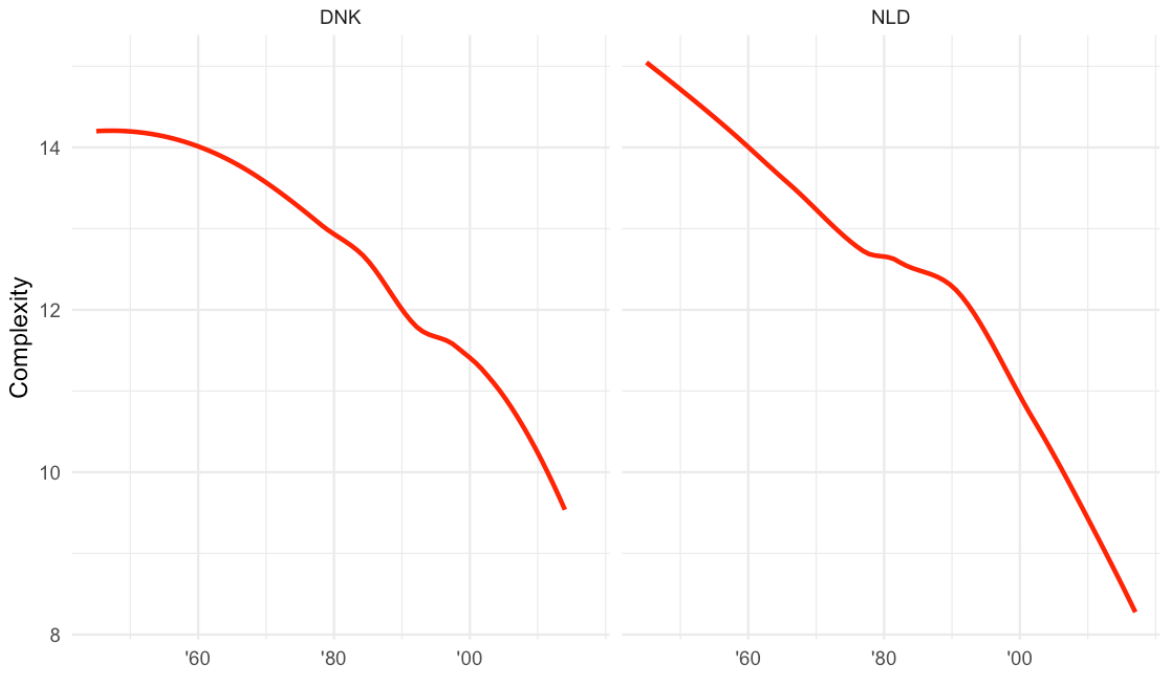
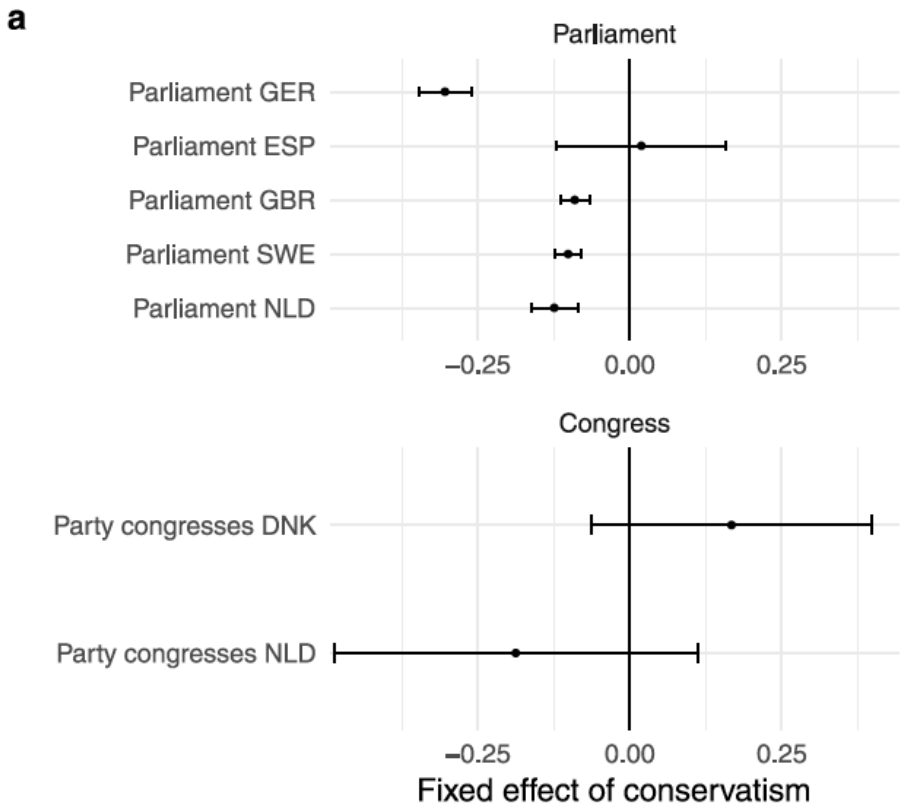


Replication

Figure 4a – OLS Regression: Complexity on Ideology (Libs-Cons) with Fixed Effects (Speaker)



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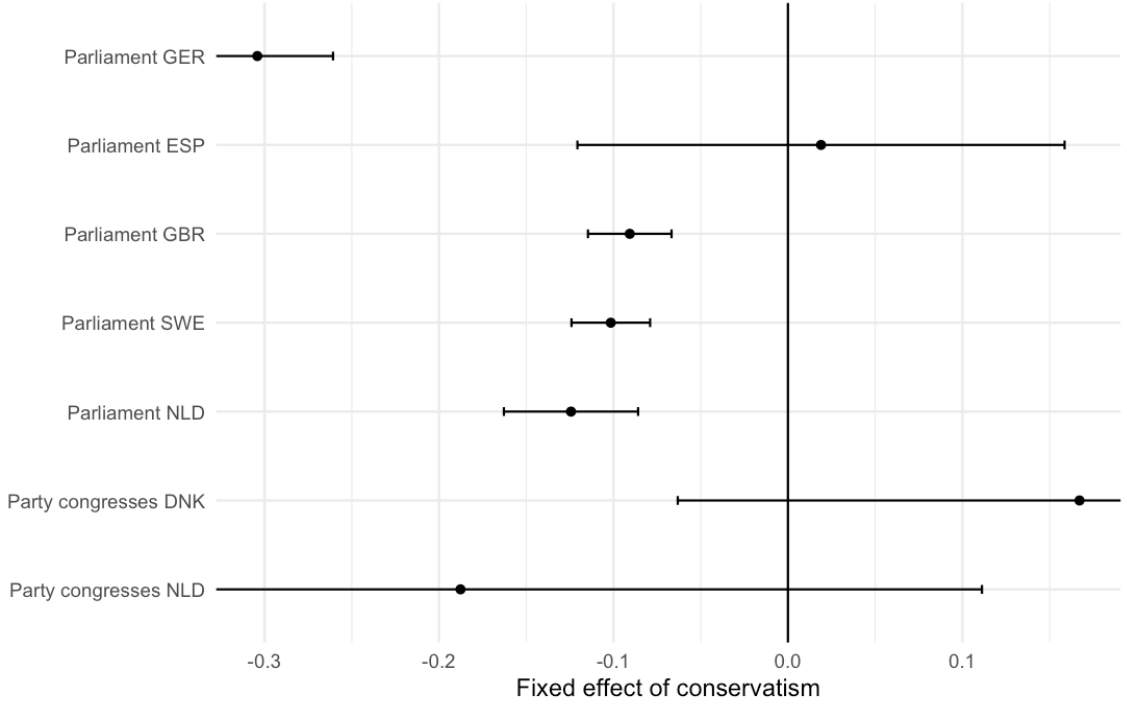
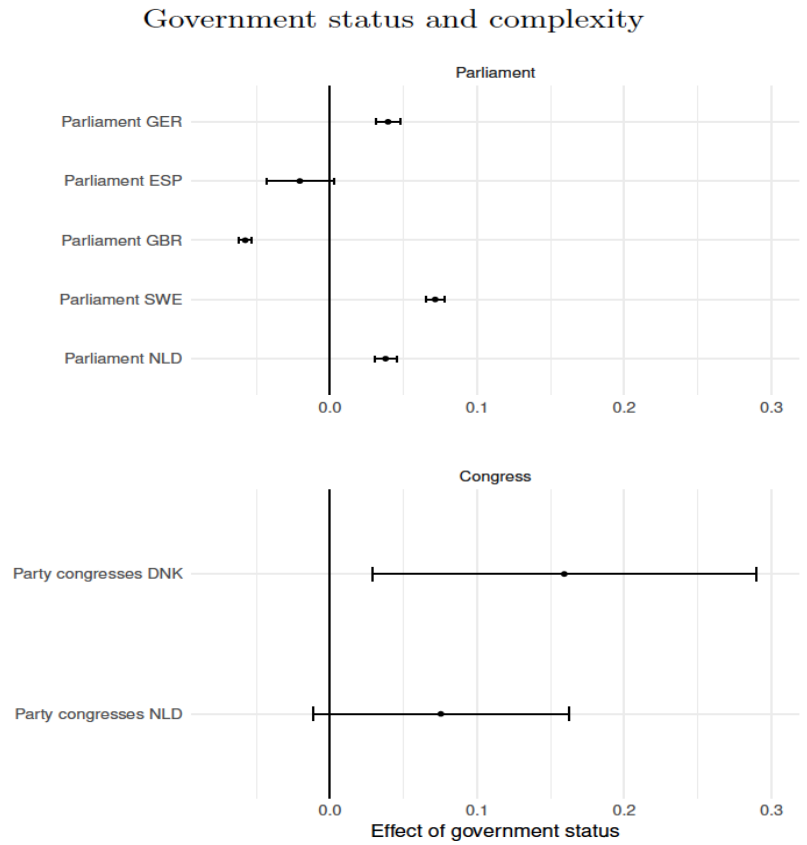


Replication

Appendix B – OLS Regression: Impact of Government vs Opposition Status



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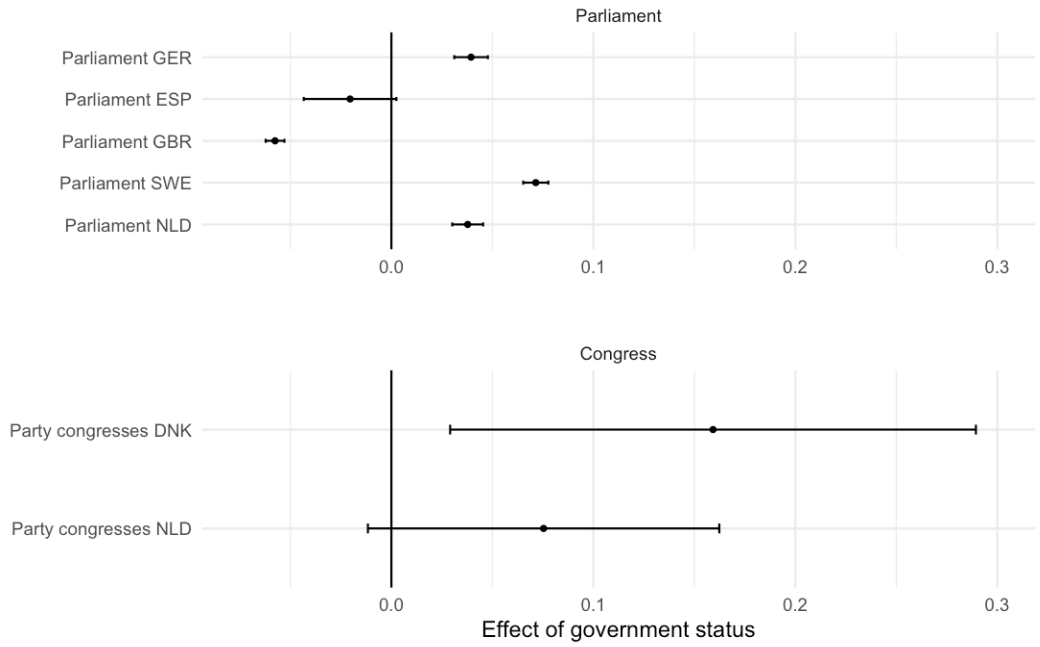


Replication

date	speaker	party	parliament	text	terms	length	sentences	sentlength	termlength	flesh	flesh.grade	year
1991-12-12	Peter Lloyd	Con	UK-HoC	I congratulate my hon. Friend the Member for Hamps…	2789	2769	159	17.415094	4.665944	47.51708	10.958044	1991
1995-02-23	John Reid	Lab	UK-HoC	May I first join the tribute paid to our armed forces es…	6109	6079	290	20.962069	4.704557	42.48026	12.541746	1995
2009-03-26	Philip Hammond	Con	UK-HoC	The simple way to get the banks lending to business …	169	165	11	15.000000	4.963636	43.43182	10.927879	2009
2008-11-10	Gerald Kaufman	Lab	UK-HoC	It is always a great pleasure to follow the hon. Membe…	1264	1255	75	16.733333	4.660558	51.72684	10.201498	2008
2013-07-05	Gerald Howarth	Con	UK-HoC	Can the right hon. Gentleman give the House an indic…	341	339	21	16.142857	4.448378	56.68717	9.362941	2013
2012-11-27	David Nuttall	Con	UK-HoC	I beg to move amendment in page line after approved…	3409	3362	142	23.676056	4.668352	43.74979	13.038903	2012
2009-03-19	Tony Baldry	Con	UK-HoC	What recent assessment he has made of the future pr…	234	232	11	21.090909	4.987069	35.18980	13.590627	2009
2012-06-12	Rob Flello	Lab	UK-HoC	What assessment he has made of the potential effect …	3332	3324	173	19.213873	4.583634	51.37227	10.867189	2012
2010-12-01	Edward Leigh	Con	UK-HoC	The sitting is resumed. Order. Five Back Benchers hav…	73	72	11	6.545455	4.444444	70.94136	4.990505	2010
2009-06-11	Paul Holmes	LibDem	UK-HoC	I congratulate the Chairman of the Home Affairs Com…	7108	7055	325	21.707692	4.760737	41.76714	12.826446	2009
2010-10-27	Alan Reid	LibDem	UK-HoC	I am sure that the Secretary of State will be aware of t…	1254	1250	62	20.161290	4.716800	43.49881	12.200743	2010
2003-04-03	Colin Breed	LibDem	UK-HoC	I do not want to detain the House for too long but I w…	1607	1596	68	23.470588	4.995614	31.67589	14.671925	2003
2012-04-16	Julian Smith	Con	UK-HoC	What steps he is taking to improve the quality of voca…	227	226	16	14.125000	5.106195	44.63529	10.542644	2012
2003-09-08	Terry Lewis	Lab	UK-HoC	We have reached the stage in the Second Reading deb…	786	781	45	17.355556	4.668374	50.34946	10.548193	2003
1998-06-02	Alan Beith	LibDem	UK-HoC	We sought to amend that closed list system but does …	234	232	13	17.846154	4.590517	50.15219	10.697586	1998
1994-02-10	David Lidington	Con	UK-HoC	I agreed with the tone and substance of the speech of…	1174	1170	66	17.727273	4.739316	47.26336	11.070987	1994
1996-07-23	Iain Sproat	Con	UK-HoC	I am pleased to respond to the motion and to set out …	6631	6482	301	21.534884	4.974391	31.47808	14.218632	1996
2012-05-23	Paul Blomfield	Lab	UK-HoC	The hon. Member for Dewsbury Simon Reevell made …	1094	1072	48	22.333333	5.037313	30.03999	14.617575	2012
2014-04-02	Stephen Doughty	Lab	UK-HoC	It is a pleasure to serve under your chairmanship Mrs …	1301	1289	60	21.483333	4.795190	39.58830	13.074613	2014
1990-12-05	Michael Heseltine	Con	UK-HoC	I beg to move to leave out from House to the end of t…	3790	3764	232	16.224138	4.746812	45.73387	10.910899	1990
2014-05-13	Annette Brooke	LibDem	UK-HoC	How big a contribution is the Governments universal f…	2522	2476	113	21.911504	4.676090	42.49006	12.776246	2014
2000-01-19	Jeremy Corbyn	Lab	UK-HoC	I welcome the opportunity to hold this brief debate. I …	2933	2920	147	19.863946	4.849658	39.98481	12.617007	2000
2006-01-25	Mark Pritchard	Con	UK-HoC	My hon. Friend is generous to give way to so many ho…	828	823	52	15.826923	4.675577	50.96994	10.081892	2006
2010-11-03	Rehman Chishti	Con	UK-HoC	I welcome the Minister's statement as a constructive a…	234	234	12	19.500000	5.038462	40.98096	12.387650	2010
2012-05-24	Alistair Burt	Con	UK-HoC	I thank the hon. Member for Bradford South Mr Sutcli…	1831	1820	89	20.449438	4.845604	40.35300	12.711105	2012
2001-11-22	Lindsay Hoyle	Lab	UK-HoC	The new free places that have been created in nurseri…	175	174	14	12.428571	4.586207	59.05448	8.110016	2001
1999-05-18	Douglas Hogg	Con	UK-HoC	As the debate on incapacity benefit is now to be fixed…	1427	1424	91	15.648352	4.477528	55.73451	9.372970	1999

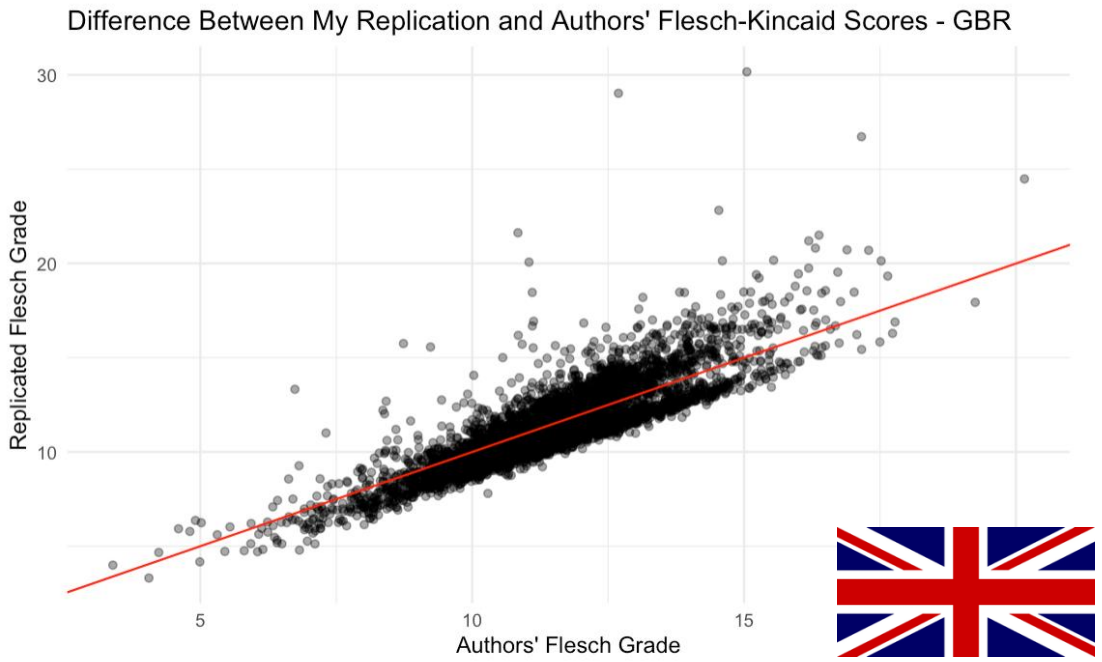
Replication: Phase 2a

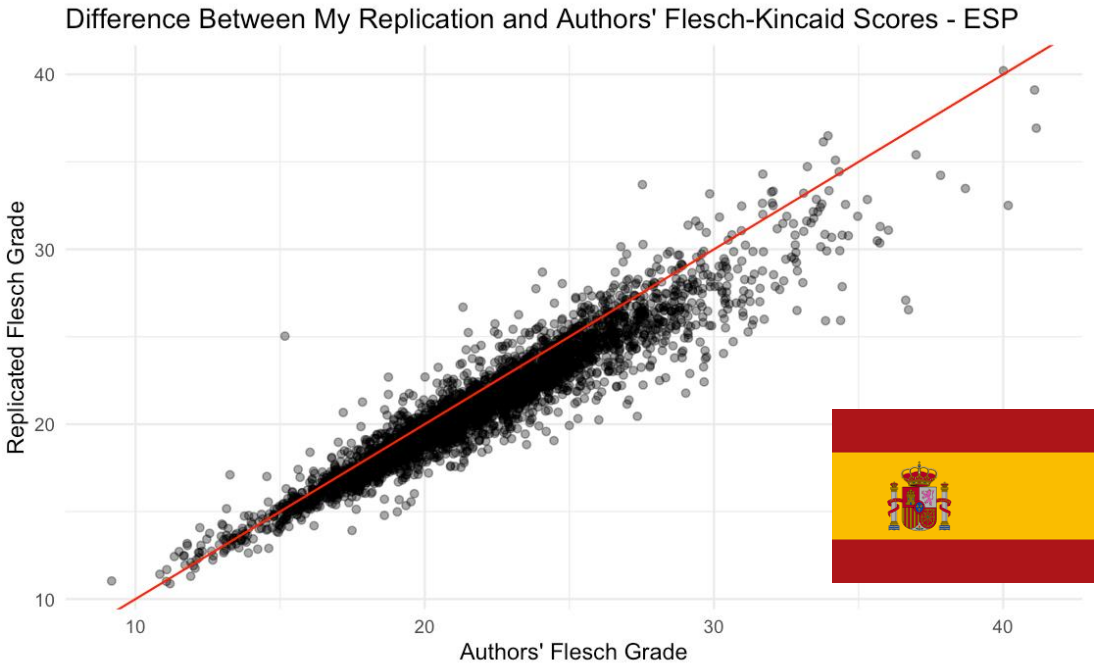
Unprocessed Data – Flesch Grade Score - Parliament

4 Corpora (Parliaments = Great Britain, Netherlands, Germany, Spain); Sweden no

4% Partial Corpora (~ 3,000 speeches per country; ~ 16,000 total speeches)

Comparing FK Scores: Replication vs Authors





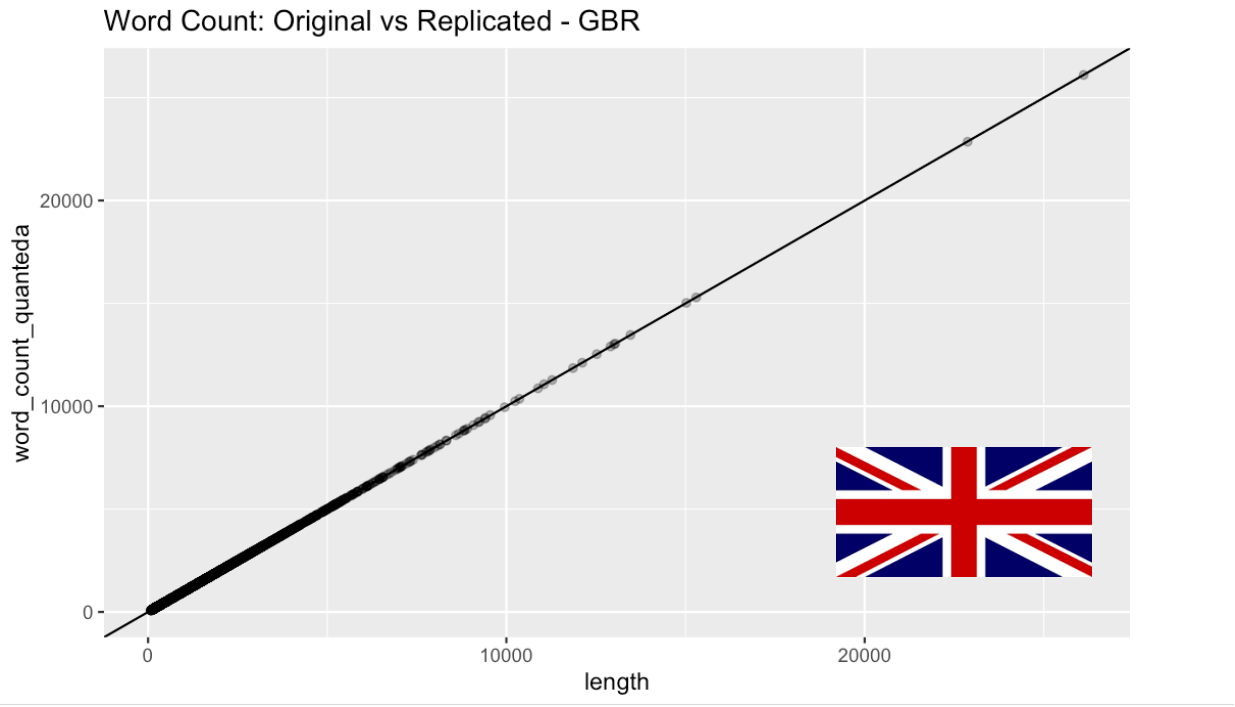
Replication:
Uses textstat_readability R(Quanteda)

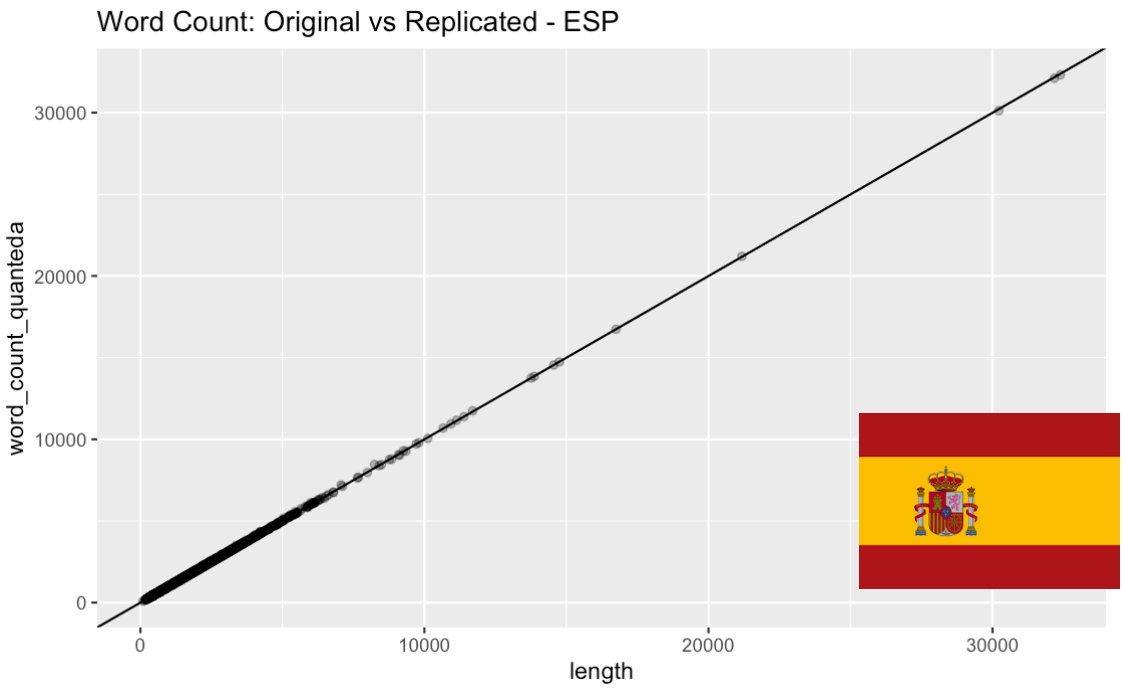
Flesch–Kincaid Grade Level

$$0.39\left(\frac{\text{total words}}{\text{total sentences}}\right)+11.8\left(\frac{\text{total syllables}}{\text{total words}}\right)-15.59$$

Authors:
Use ?

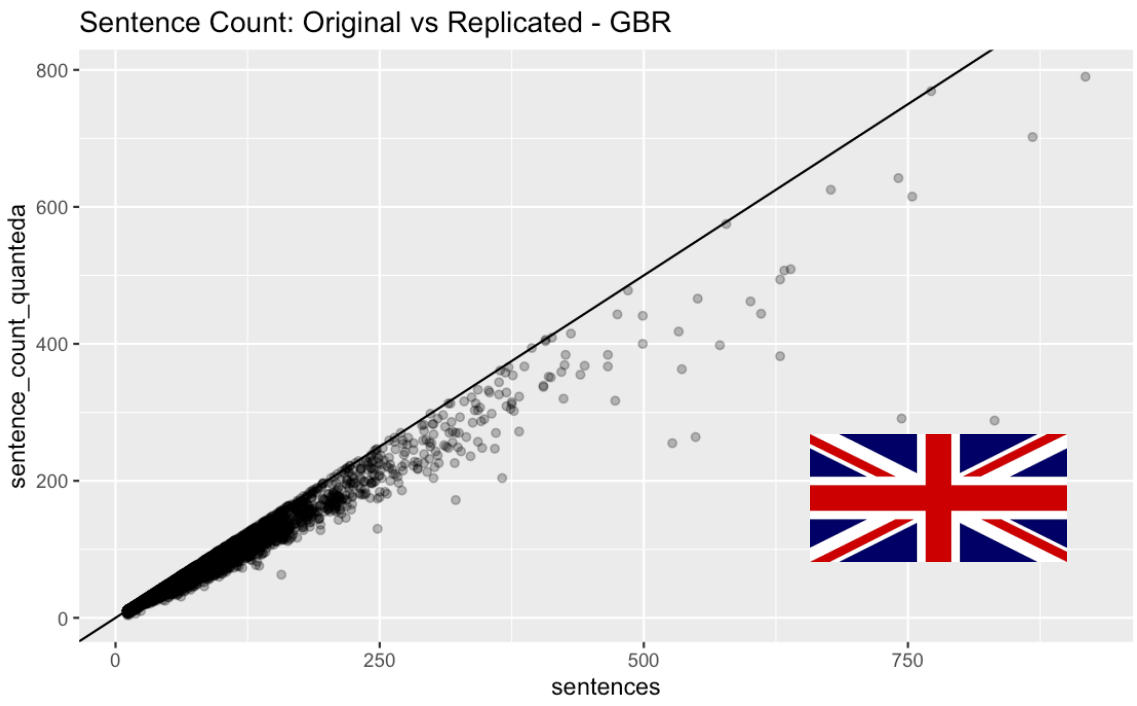
Comparing Word Counts: Replication vs Authors

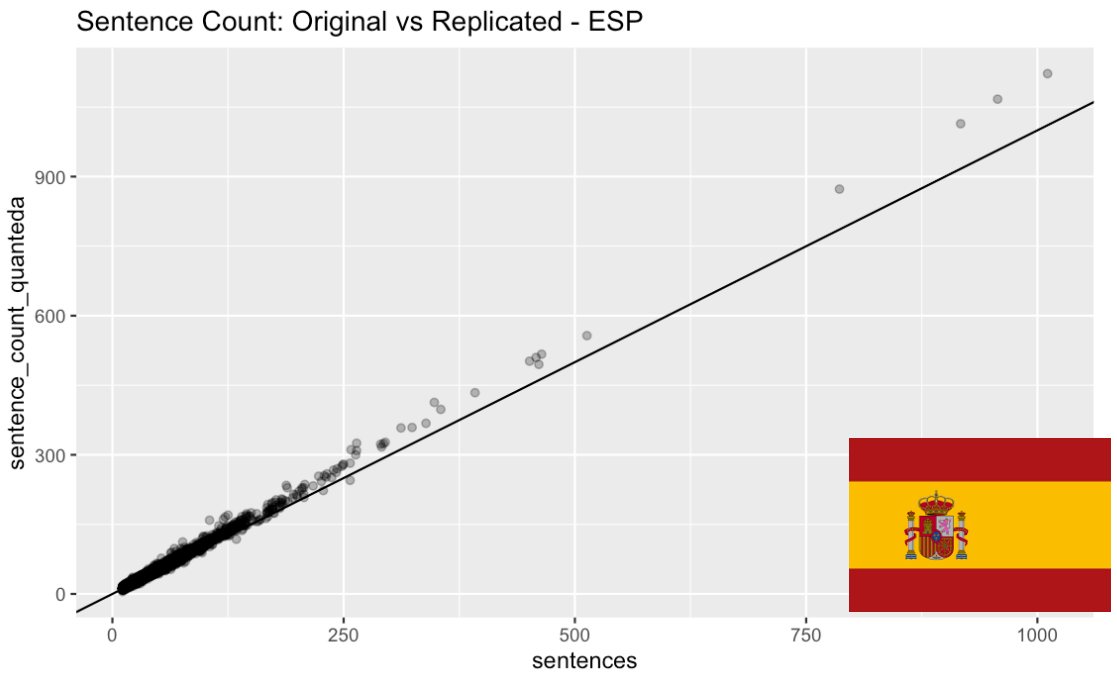




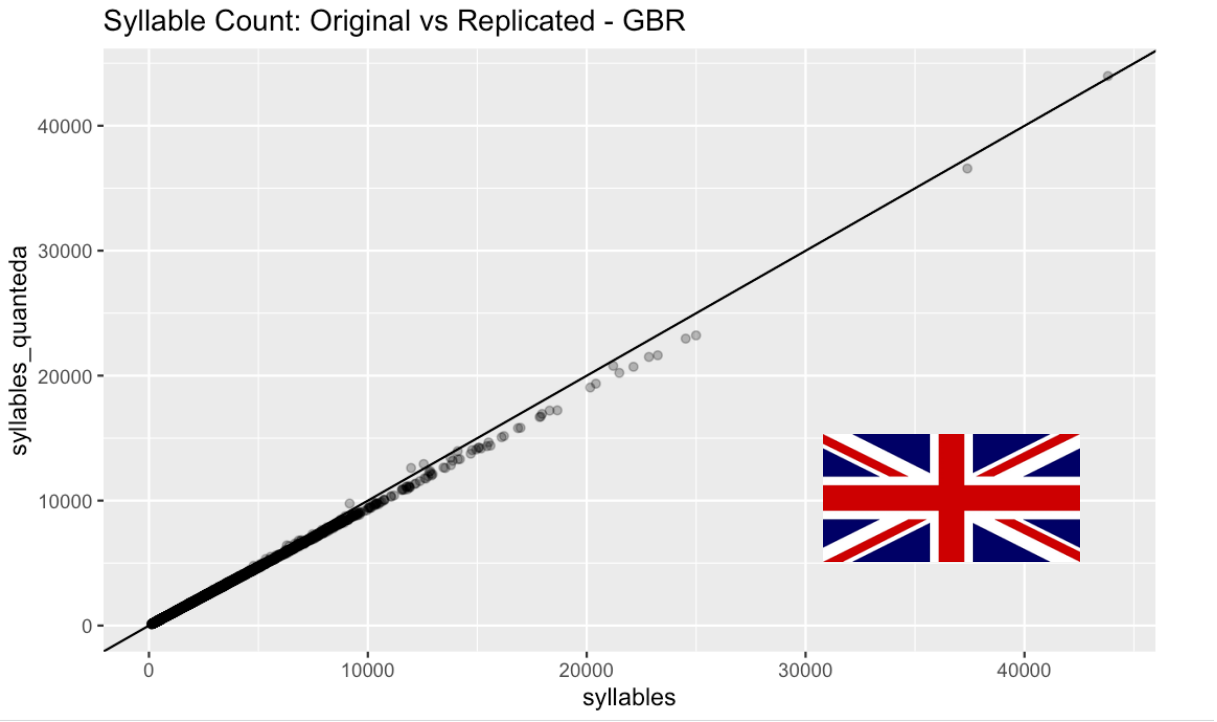
Note: Length = words

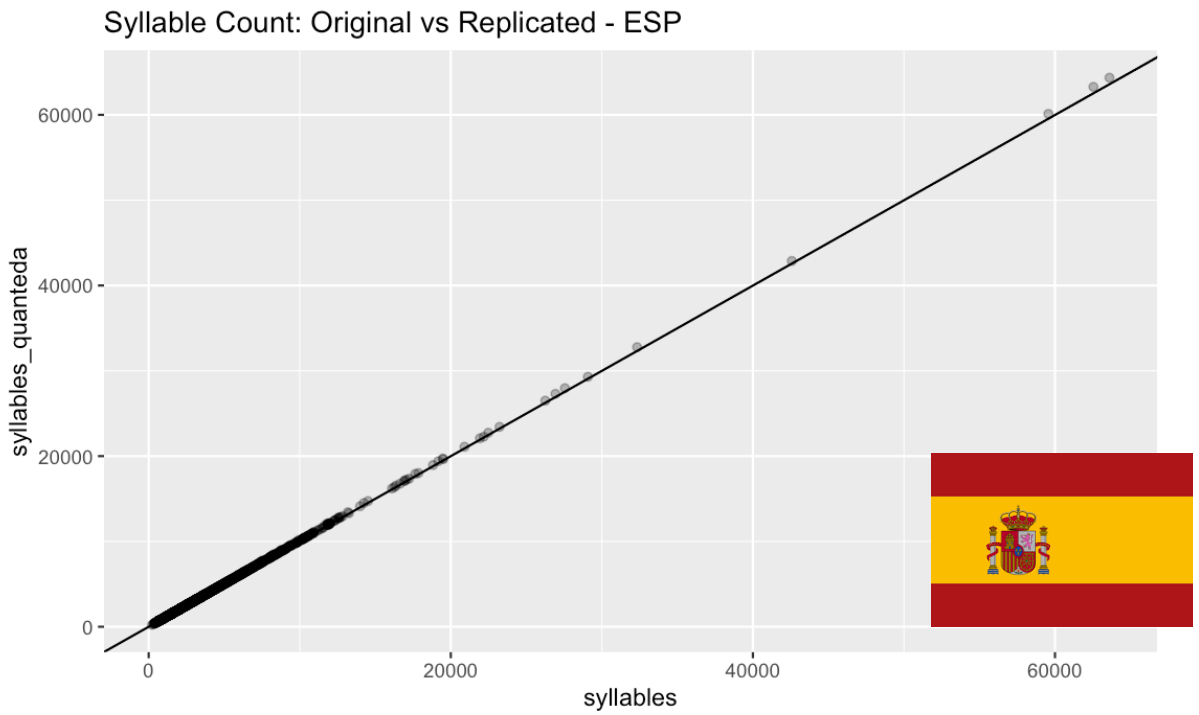
Comparing Sentence Count: Replication vs Authors



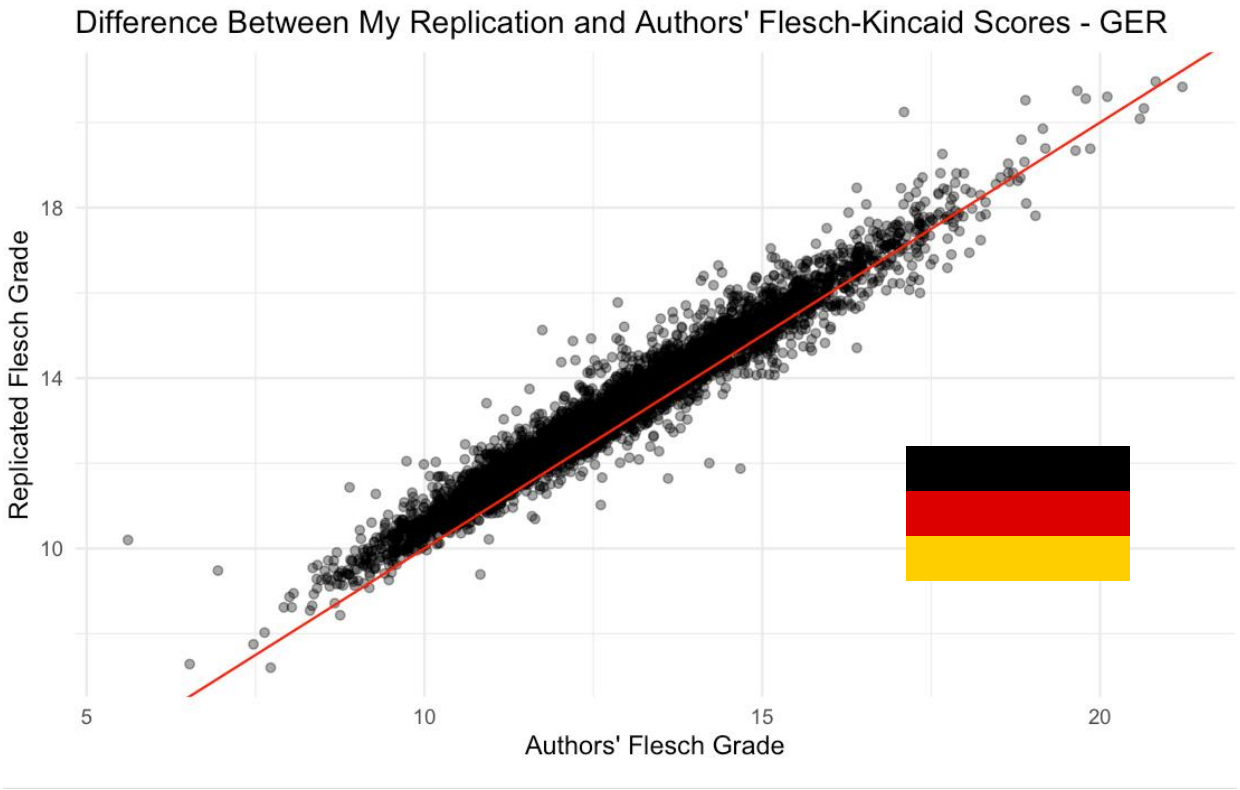


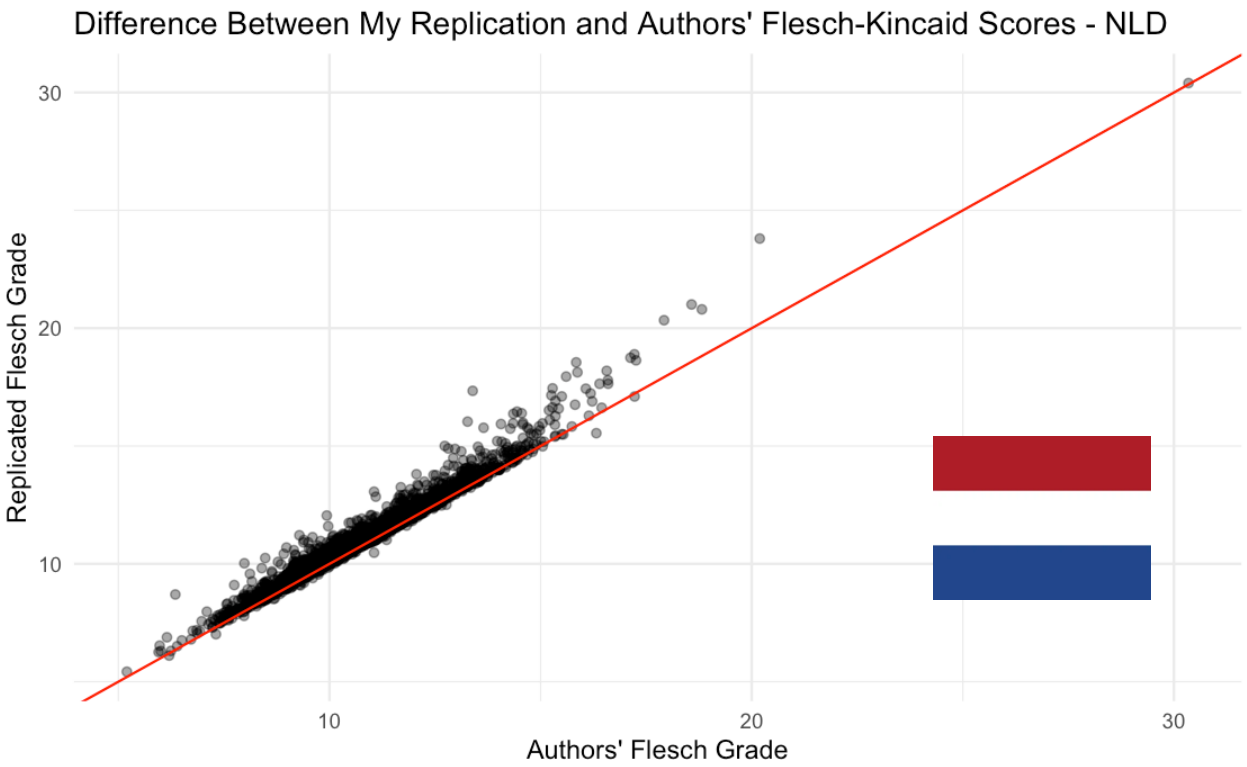
Comparing Syllable Count: Replication vs Authors



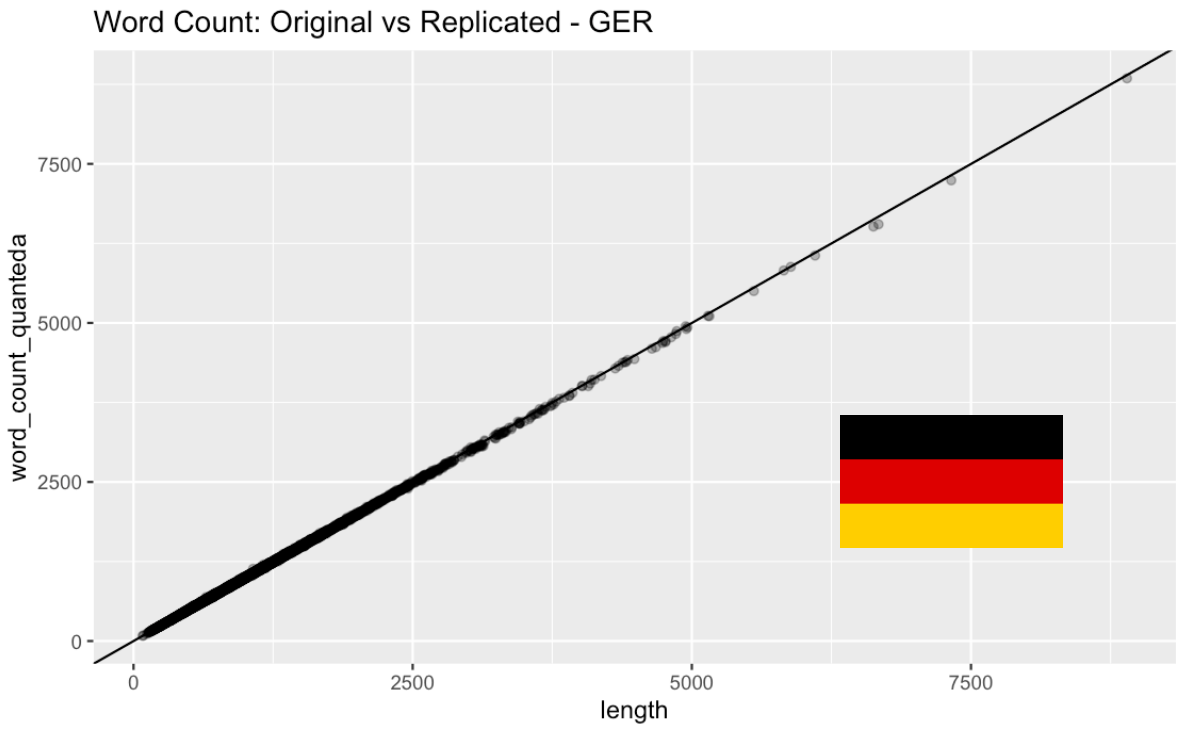


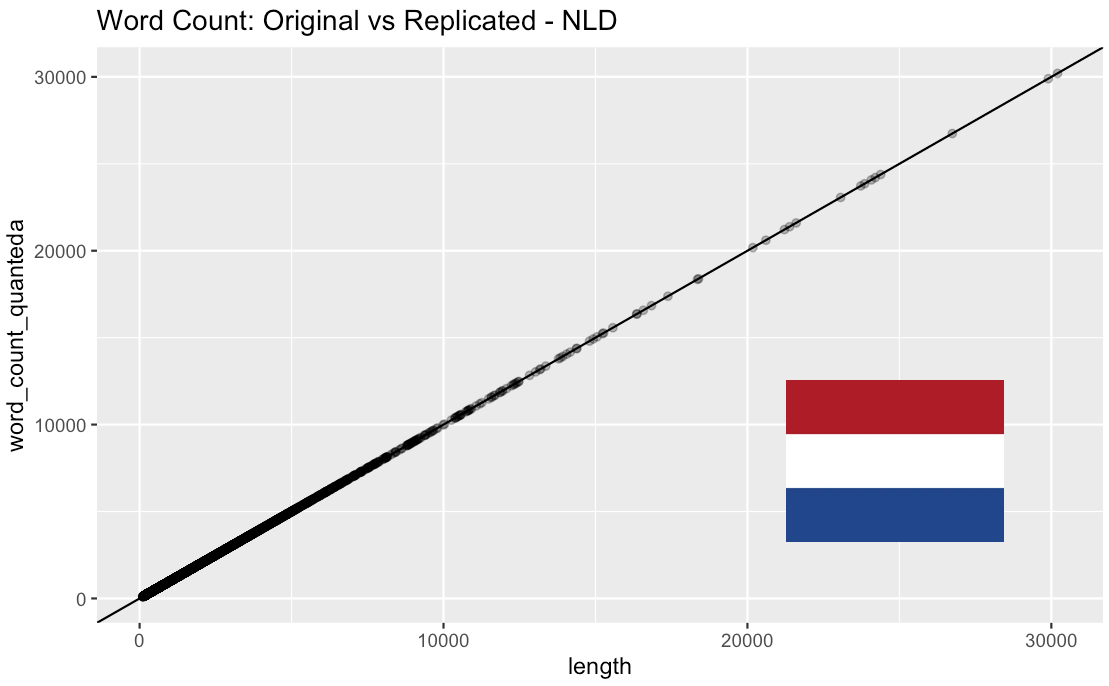
Comparing FK Scores: Replication vs Authors



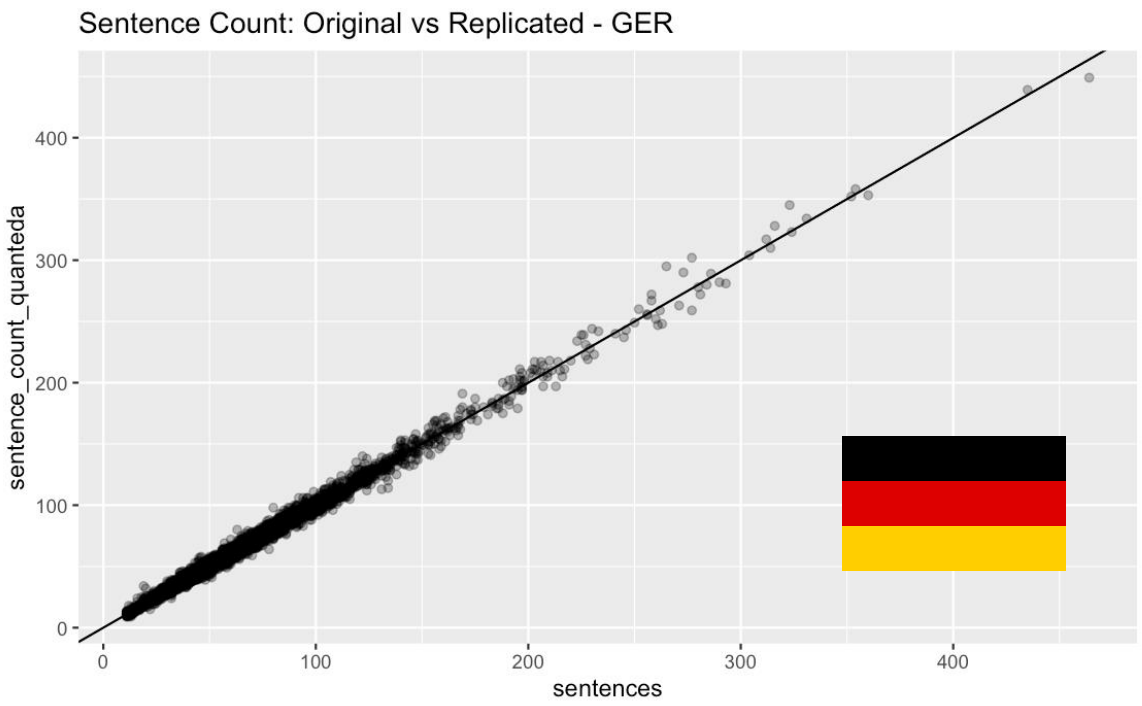


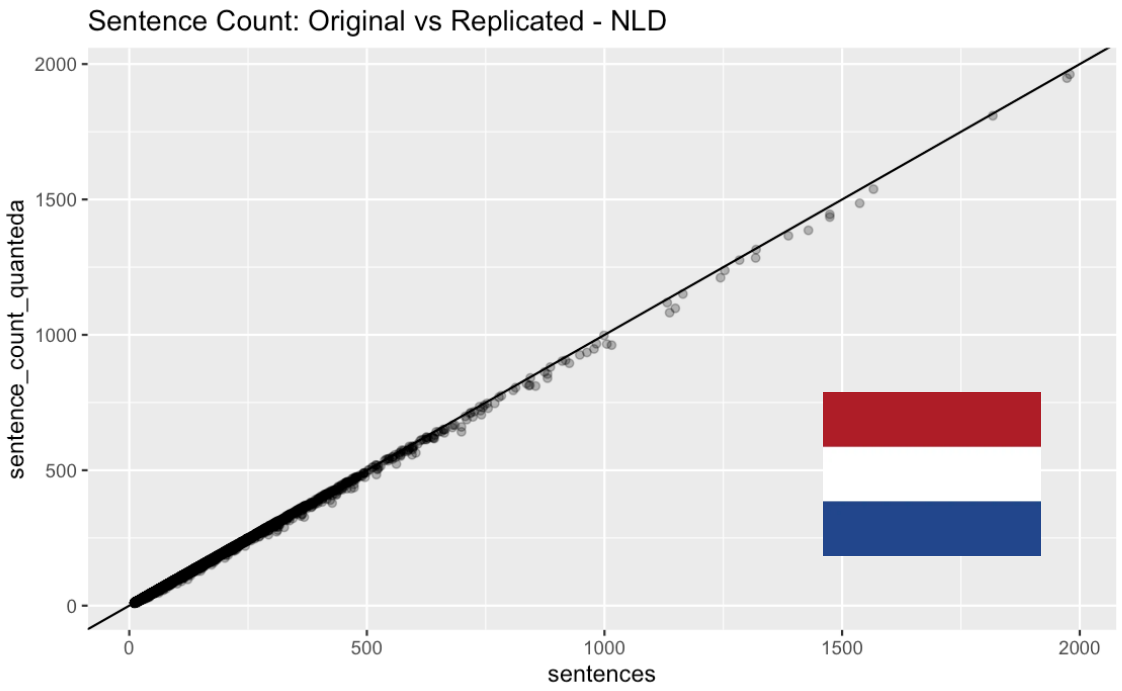
Comparing Word Counts: Replication vs Authors





Comparing Sentence Count: Replication vs Authors





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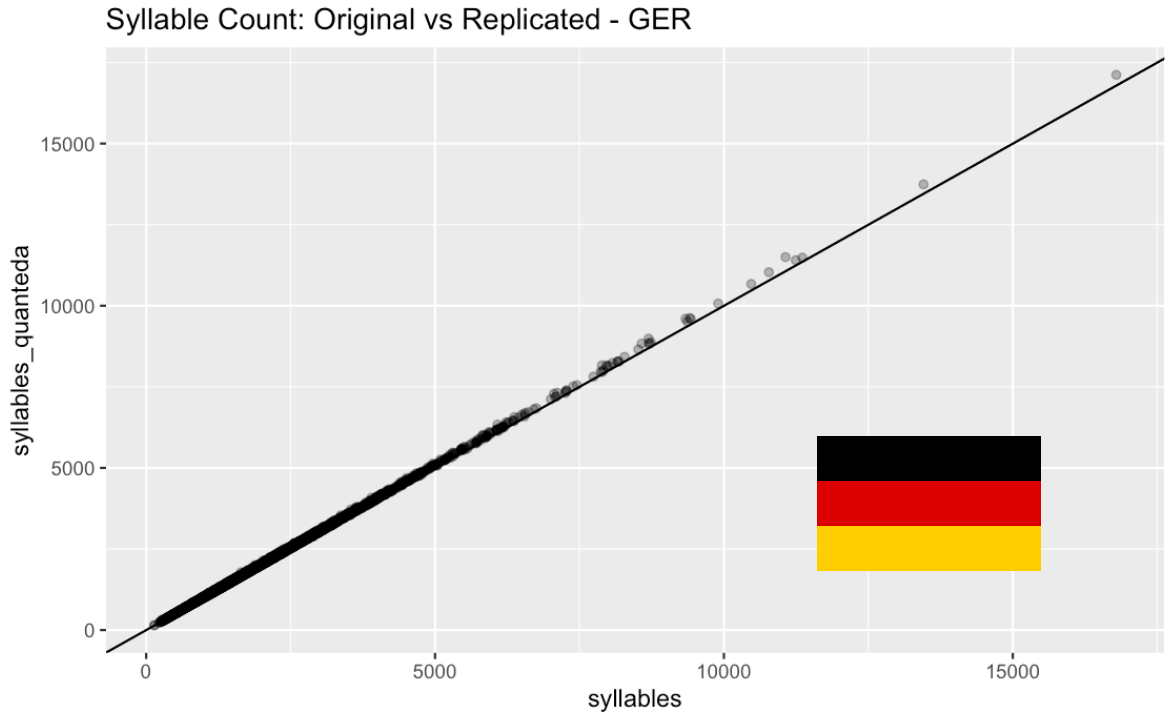


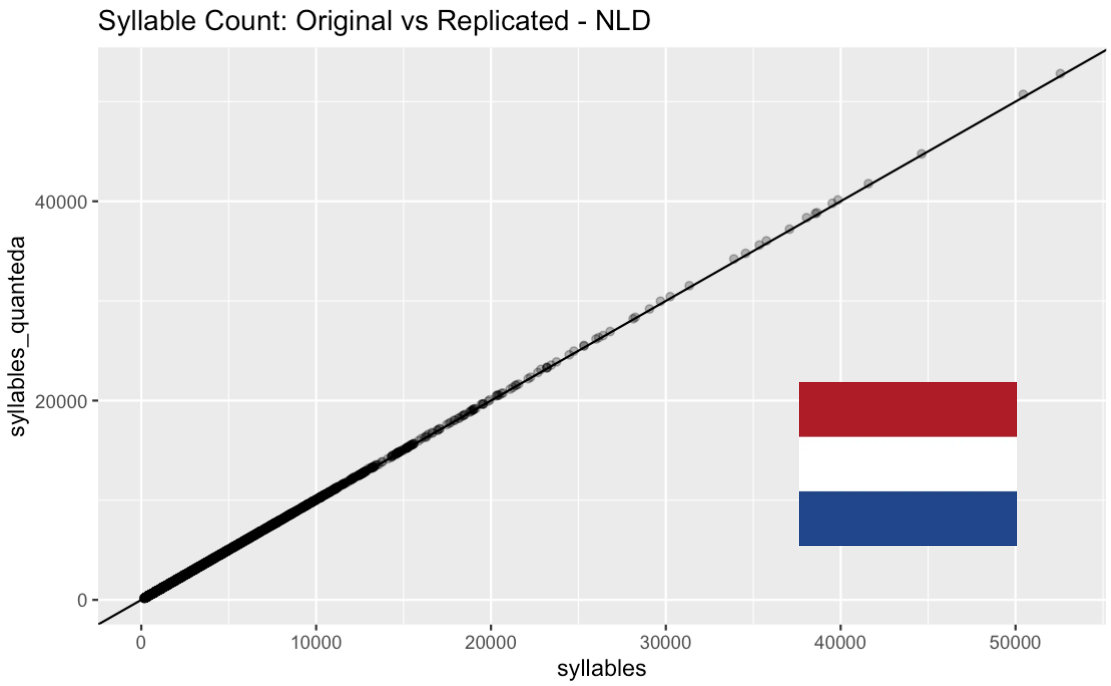


Figure 1 – Mean Complexity by Select Speaker (Parliament)

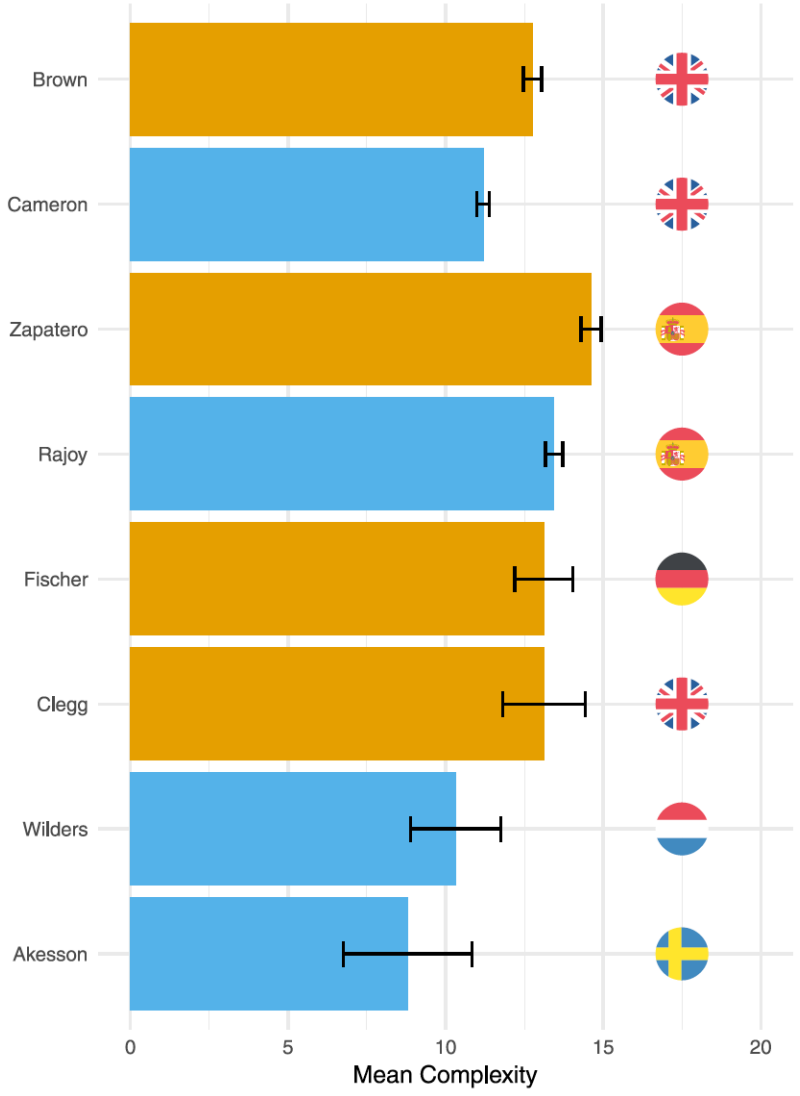
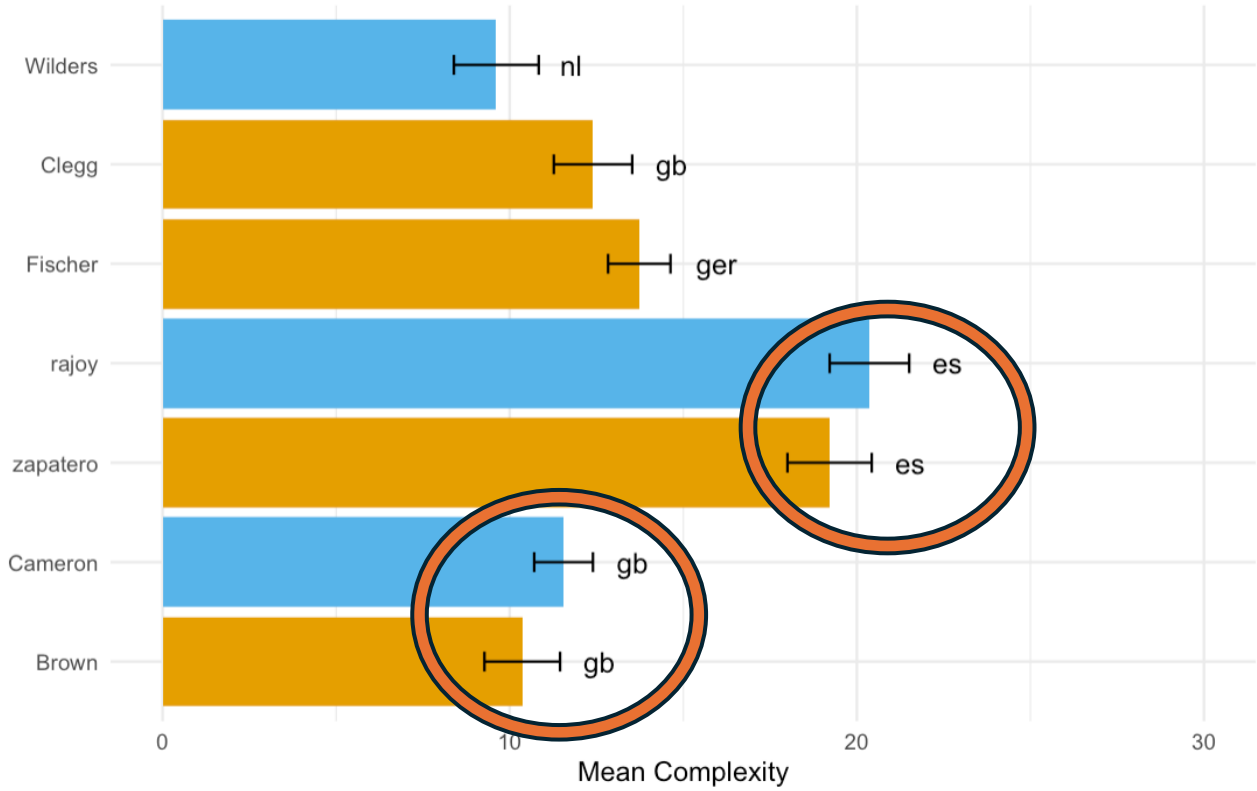
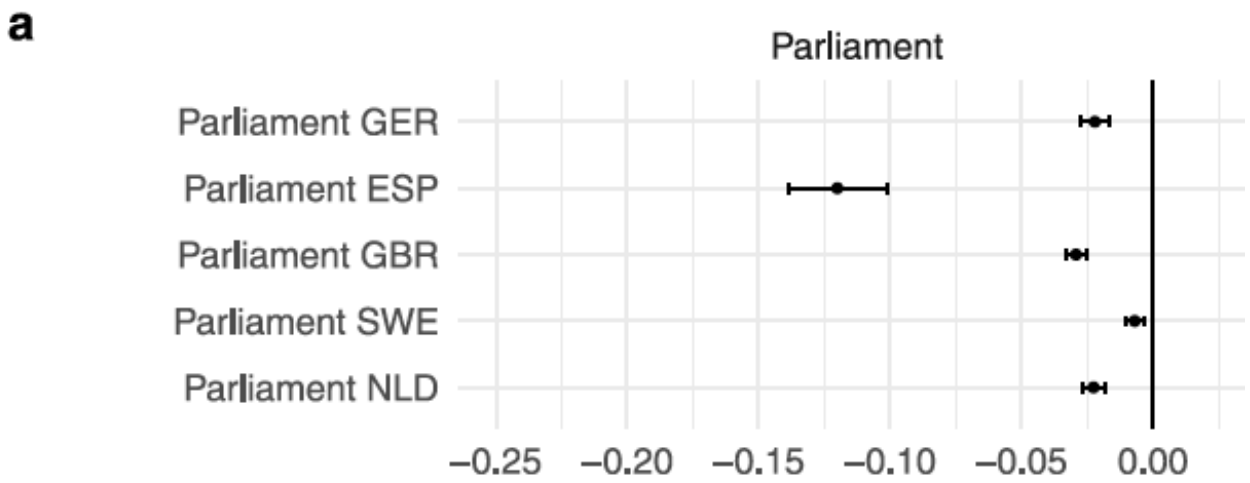


Fig 1. Descriptive information on linguistic complexity. The bars in this figure denote mean complexity scores, with 95% confidence intervals, of selected speakers.



Replication

Figure 2a – OLS Regression: Complexity on Ideology (Libs-Cons Social Ideology)



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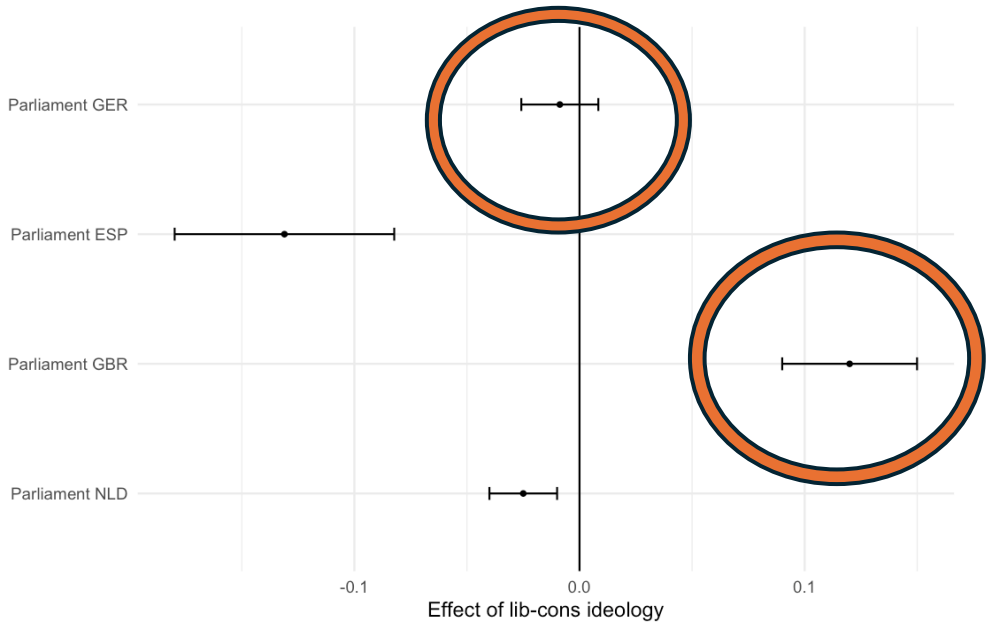
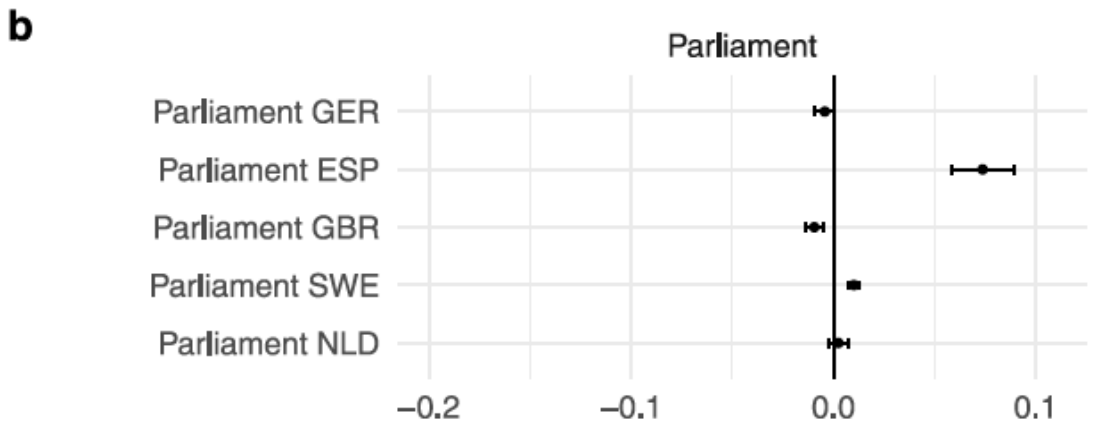


Replication

Figure 2b – OLS Regression: Complexity on Ideology (Left-Right Economic Ideology)



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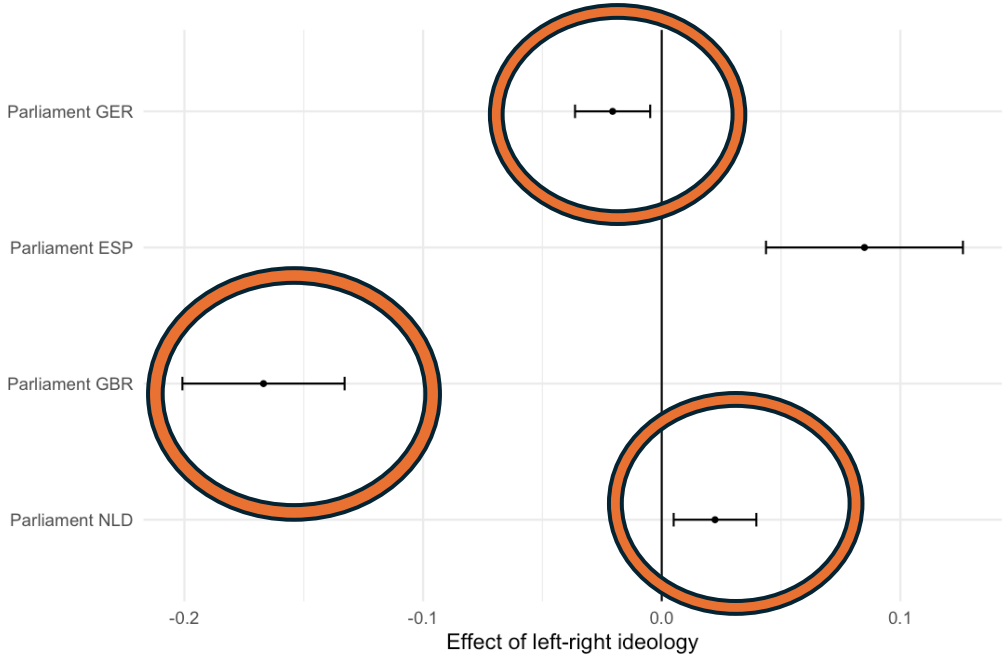
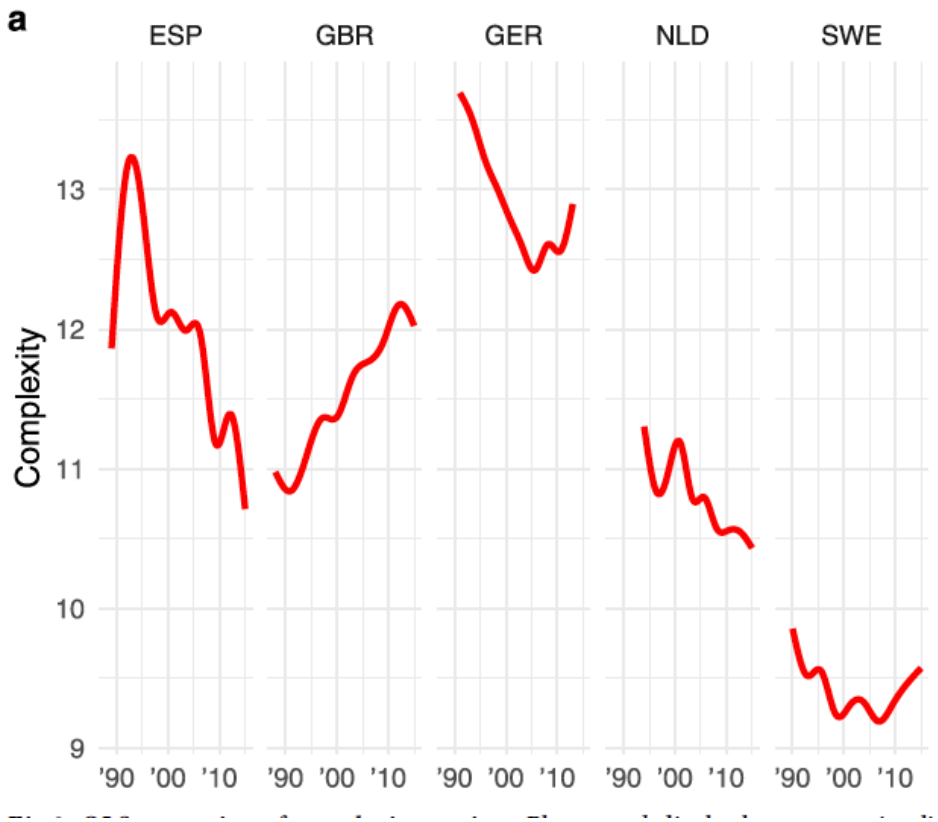


Replication

Figure 3 – OLS Regression: Complexity on Time



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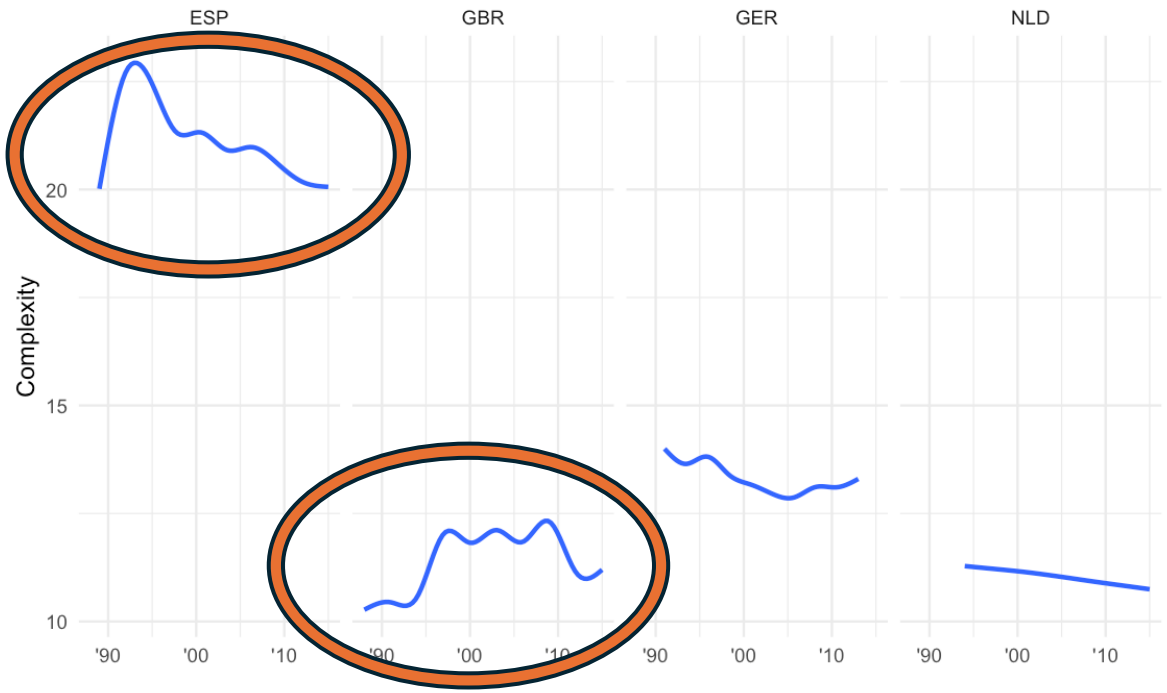
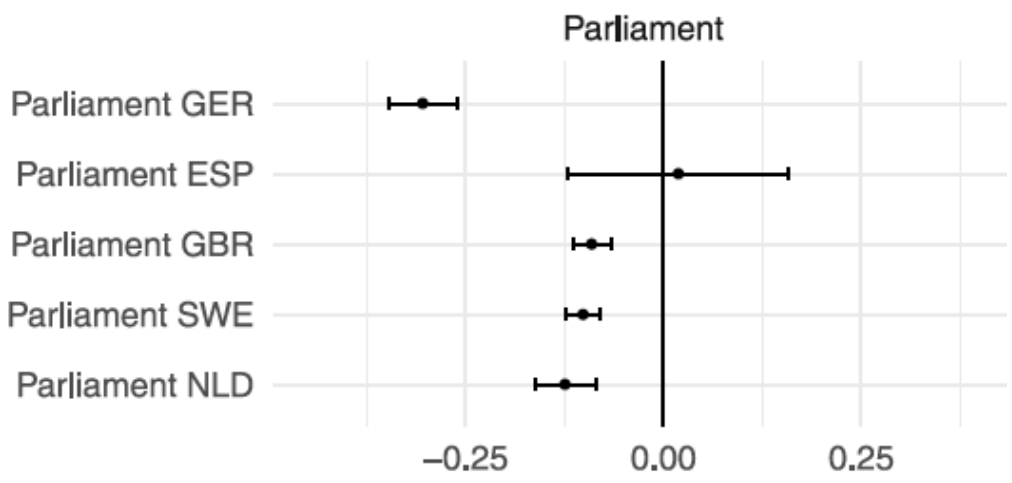


Replication

Figure 4a – OLS Regression: Complexity on Ideology (Libs-Cons) with Fixed Effects (Speaker)

a



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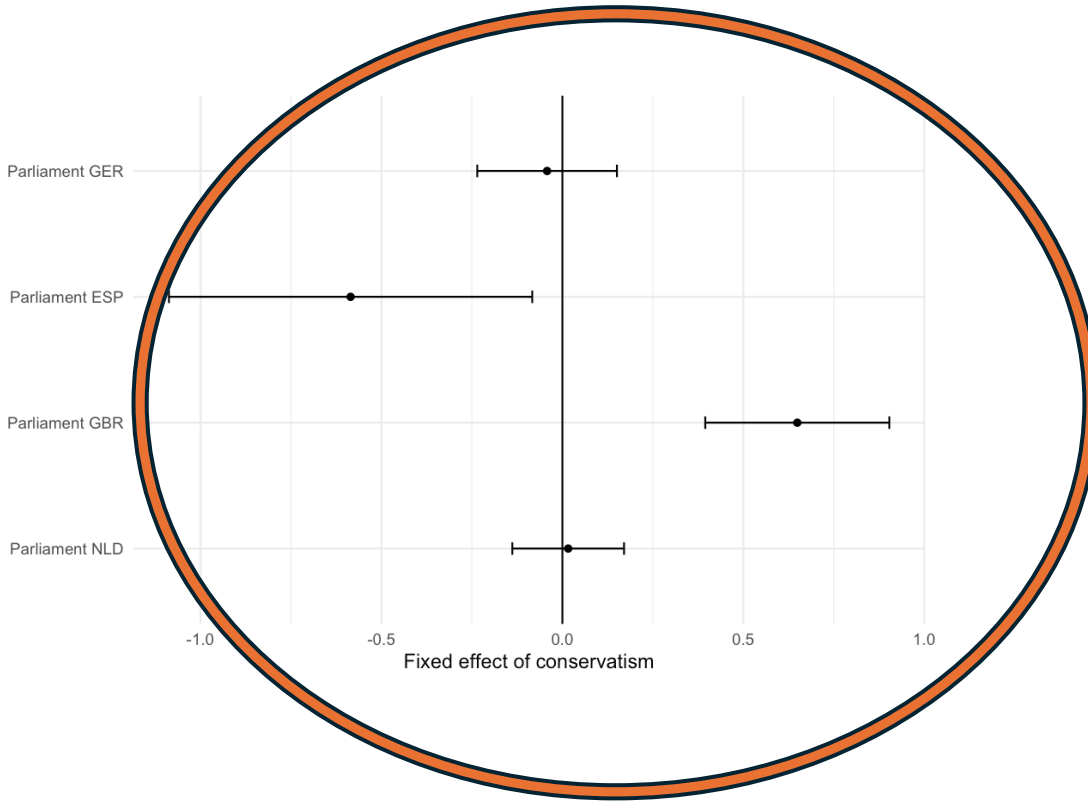
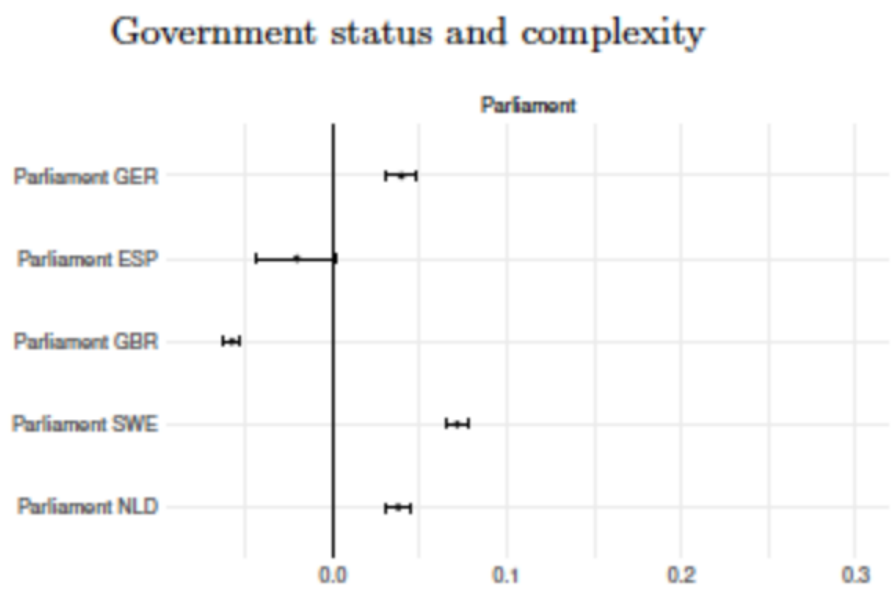


Replication

Appendix B – OLS Regression: Impact of Government vs Opposition Status



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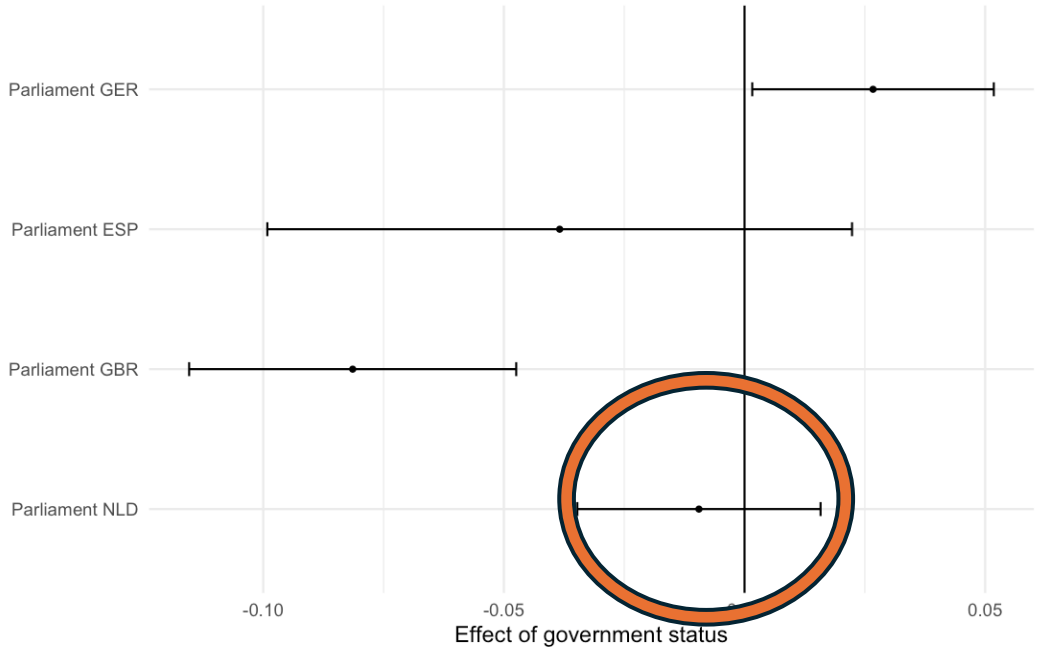


Replication

text	date	speaker	document	TTR	ARI	I	avg_sl	entropy	mtld	syntactic_depth	syntactic_dependency
Excellencies, Ladies and Gentlemen, Distinguished De…	2015-11-30	B. Sobotka	NA	0.4553687	15.403485	17.073722	25.354839	77.45944	97.86518	6.030303	6.242424
Ladies and Gentlemen, 17 November is a day that has…	2015-11-17	B. Sobotka	NA	0.4780427	12.421207	19.231717	22.444444	77.18279	95.33391	10.400000	10.400000
British Premier David Cameron introduced today his l…	2015-11-10	B. Sobotka	NA	0.5165165	15.217910	13.766403	24.071429	78.67026	74.53135	6.279070	6.302326
I am glad that the European Commission, in the prop…	2015-09-09	B. Sobotka	NA	0.5777126	16.356228	25.872667	28.500000	81.70904	95.81926	5.488550	5.908397
On Monday 24 August 2015, Prime Minister Bohuslav …	2015-08-24	B. Sobotka	NA	0.3275296	13.017484	7.475157	21.368056	69.03344	80.17732	7.941176	7.176471
In his address, the Prime Minister stressed that energ…	2015-07-14	B. Sobotka	NA	0.3354577	12.667264	7.238594	21.812500	68.97052	76.22492	6.177215	6.392405
On Tuesday 30 June 2015, Prime Minister Bohuslav S…	2015-06-30	B. Sobotka	NA	0.4318182	13.635443	15.353636	26.290323	75.98860	92.26118	4.384365	5.221498
The Czech Republic, as well as the EU, is aware of the…	2015-05-13	B. Sobotka	NA	0.4731183	14.120694	14.617450	25.590909	76.96052	92.12584	5.634006	6.253602
Ladies and Gentlemen, In 1989, I was one of the stud…	2015-05-06	B. Sobotka	NA	0.1957665	13.369489	3.187896	24.468013	63.22170	78.76505	7.750000	8.583333
Ladies and Gentlemen, In today's gala evening, we ar…	2015-05-05	B. Sobotka	NA	0.4362851	13.884341	11.476304	25.243243	73.63684	65.13448	5.729075	6.191630
Ladies and Gentlemen, The transit of the U.S. Army c…	2015-03-27	B. Sobotka	NA	0.4210526	11.965472	12.189160	19.585366	74.61972	70.37619	6.214286	5.821429
Ladies and Gentlemen, Your Excellencies, I am very pl…	2015-03-24	B. Sobotka	NA	0.3886179	14.754745	8.256866	26.913043	73.56088	61.38542	4.480000	4.986667
Chairman of the Federation of Korean Industries, Pres…	2015-02-25	B. Sobotka	NA	0.4368794	17.588697	14.068515	29.750000	75.43394	86.22222	5.333333	5.761905
Minister, Ambassadors, Ladies and Gentlemen, This is…	2014-08-25	B. Sobotka	NA	0.2755070	14.019984	4.706448	24.416667	66.60886	70.13053	6.500000	6.500000
It is a genuine pleasure for me to attend this business…	2014-04-30	B. Sobotka	NA	0.4189627	15.146151	14.399020	25.346939	74.09373	82.33239	6.000000	7.428571
Dear guests from China and the Czech Republic, per…	2013-11-13	B. Sobotka	NA	0.3120185	18.161306	6.156055	32.121951	68.56598	72.01100	7.480000	7.400000
Mr. Chairmant, May I first congratulate you on your el…	2013-09-27	B. Sobotka	NA	0.3771063	13.930293	8.762972	22.000000	71.75215	79.04189	4.698113	5.320755
Madame Chairman, Excellencies, Distinguished Collea…	2013-09-26	B. Sobotka	NA	0.4816667	13.525427	14.975973	18.666667	77.27701	109.98395	4.647059	5.470588
Dear EUCYS Guests, I consider it a great honour for th…	2013-09-21	B. Sobotka	NA	0.5917874	13.272085	26.737194	19.761905	80.04599	85.84634	5.163043	5.586957
Mr. President, Dear Minister Kohout, Ladies and Gentl…	2013-09-02	B. Sobotka	NA	0.5694444	13.768956	27.990749	22.842105	81.25141	114.72203	4.854545	6.290909
Concerning the negotiations: "…We were faced with a…	2015-07-23	A. Tsipras	NA	0.4461358	16.200889	12.714461	27.645161	74.57510	57.54064	4.625835	5.387528
Ladies and Gentlemen Members of Parliament, We ha…	2015-07-11	A. Tsipras	NA	0.3302428	12.781046	8.677306	22.940000	69.82856	68.44325	4.750000	5.600000
This meeting should have taken place a long time ago…	2015-07-08	A. Tsipras	NA	0.3443928	11.126449	9.741104	20.173333	71.53205	67.69581	6.027586	7.020690
Thank you very much Mr. President. Honorable Memb…	2015-07-08	A. Tsipras	NA	0.3454146	13.720164	8.291579	24.223684	70.03275	58.32121	6.041379	7.034483
Citizens of Athens, citizens of Greece. Today we are n…	2015-07-04	A. Tsipras	NA	0.3504902	6.282555	8.994502	12.014706	69.76534	39.26427	6.880597	6.343284
Ladies and gentlemen Members of Parliament, Today'…	2015-06-28	A. Tsipras	NA	0.2651496	12.276336	5.162250	22.092391	66.38992	62.95475	5.962500	6.050000
" Now is the time for the leading European powers to …	2015-06-27	A. Tsipras	NA	0.4145729	14.443374	10.750247	23.441176	73.45133	61.99344	4.329167	4.966667

Replication: Phase 2b

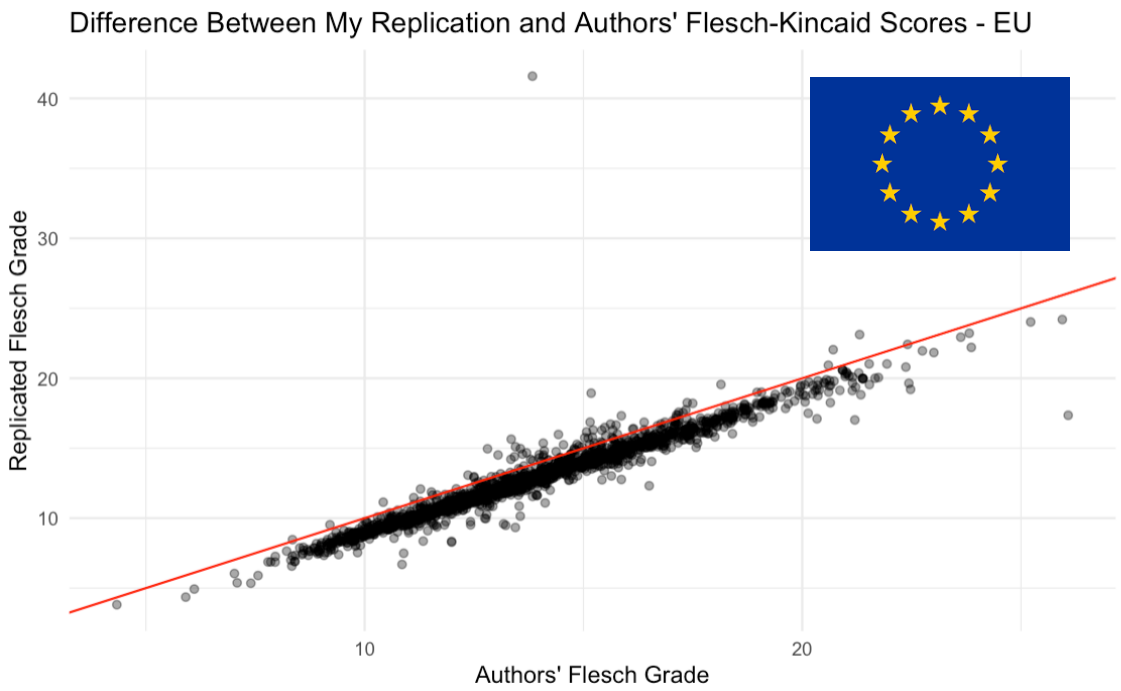
Unprocessed Data – Flesch Grade Score – EU and Congress Speeches

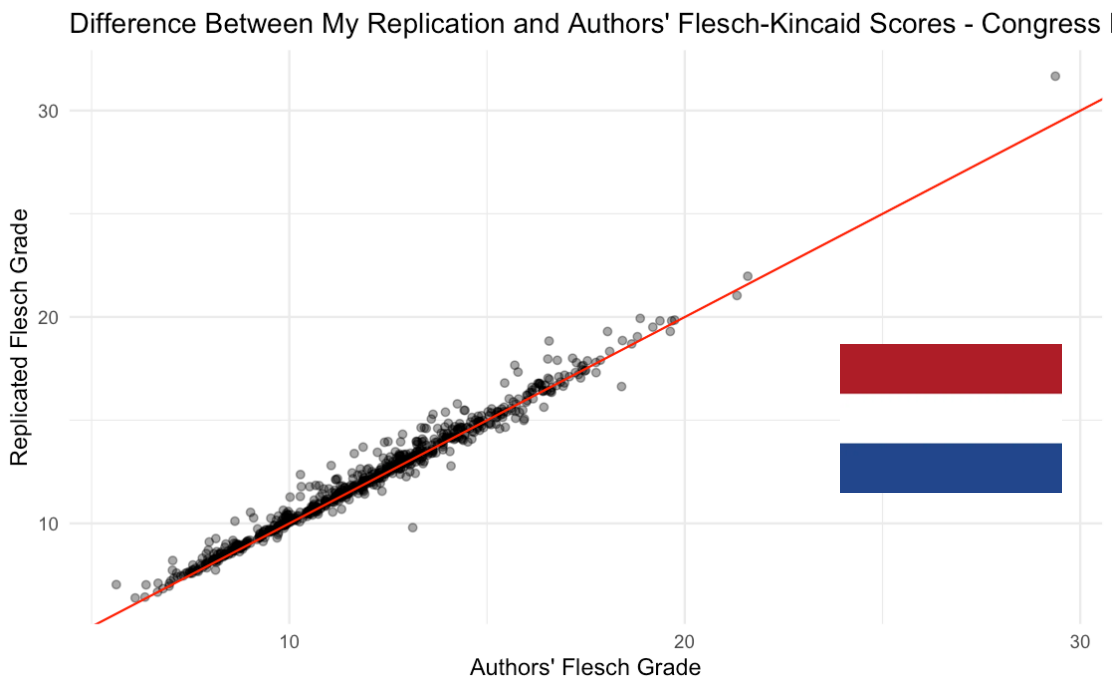
2 Corpora (EU and Congress Speeches - Netherlands)

Full Corpora (~ 1,800 speeches EU)

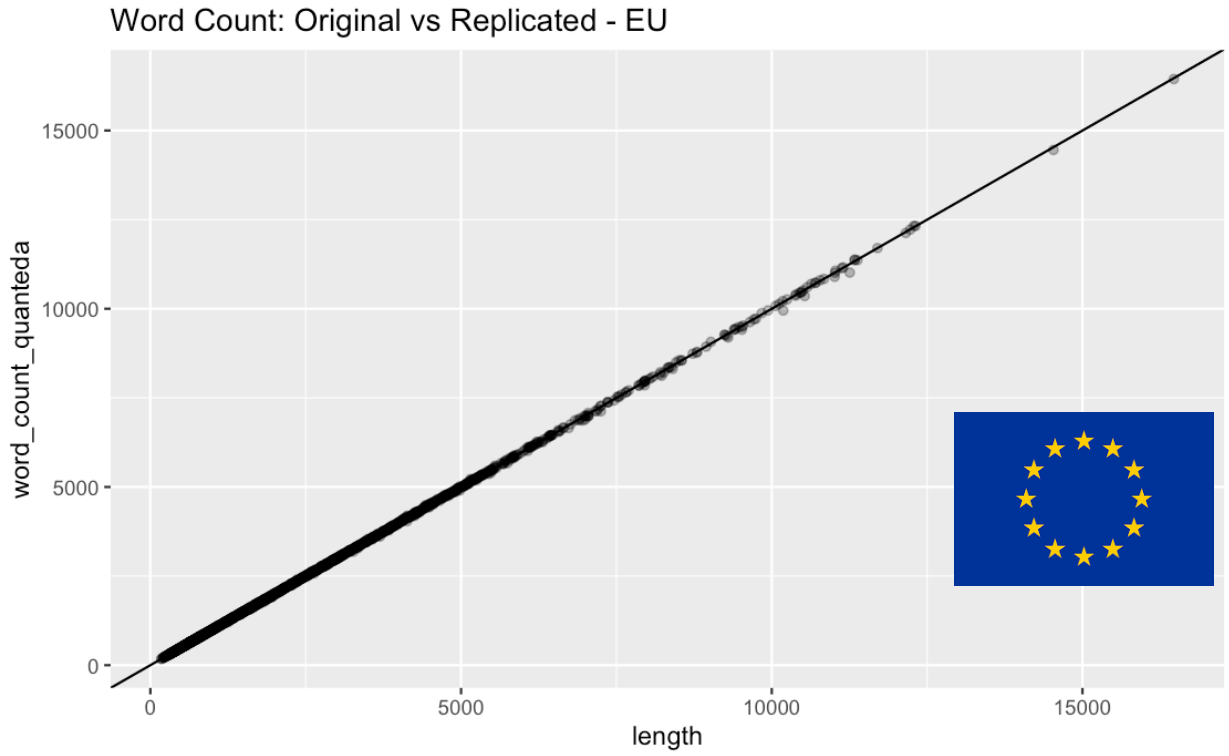
Full Corpora (~ 700 speeches, Congress NLD); Congress Denmark missing

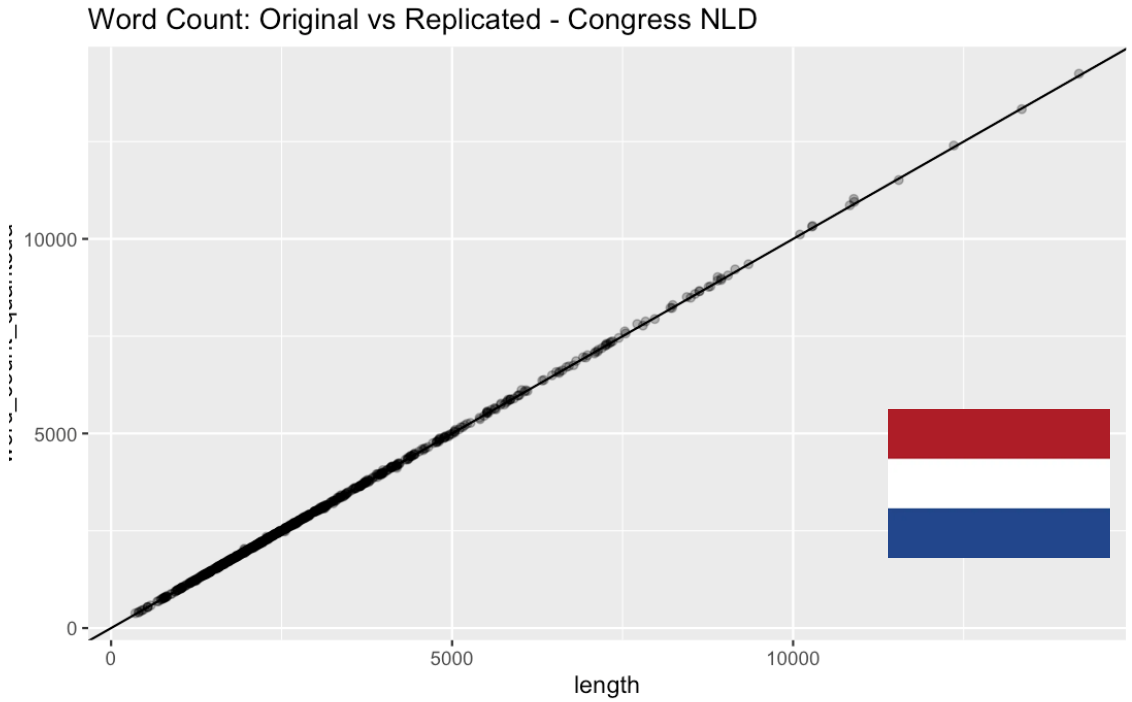
Comparing FK Scores: Replication vs Authors



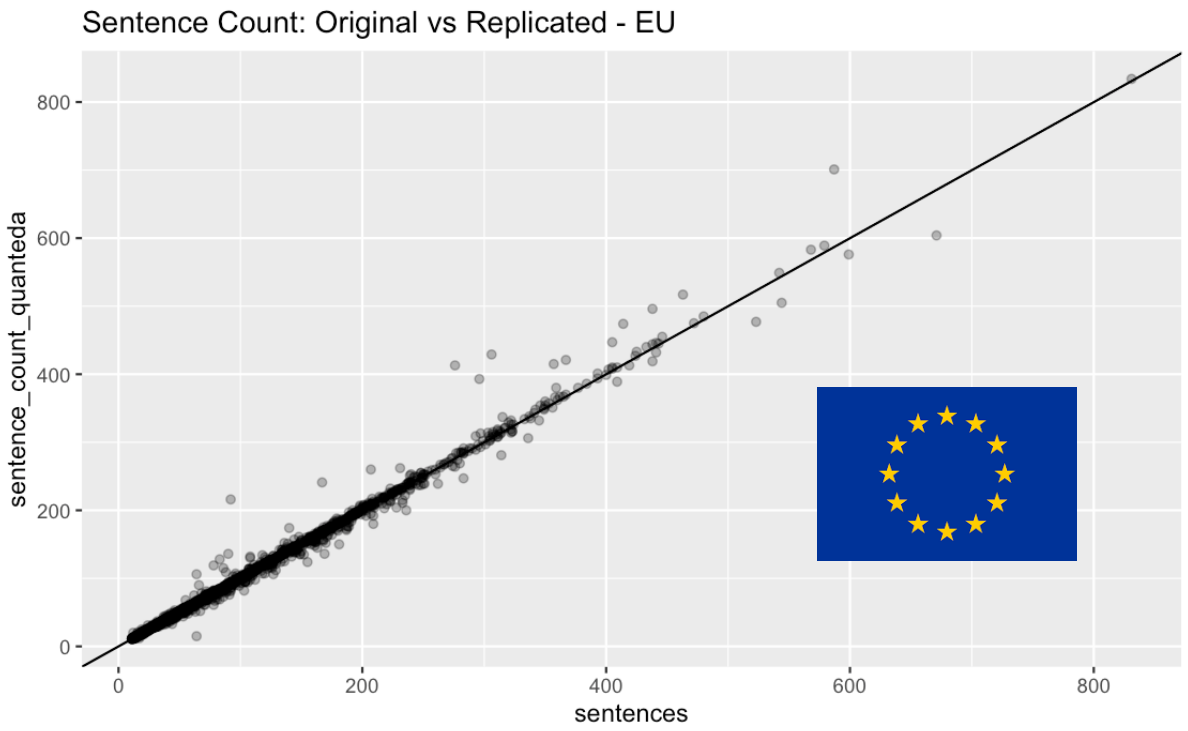


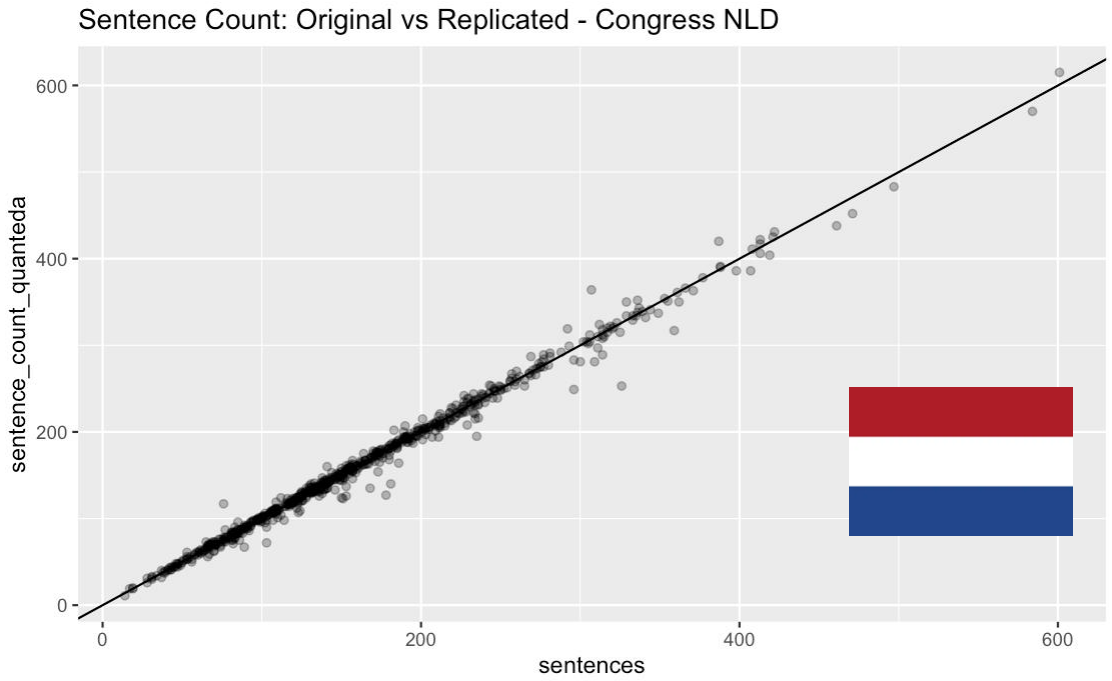
Comparing Word Counts: Replication vs Authors



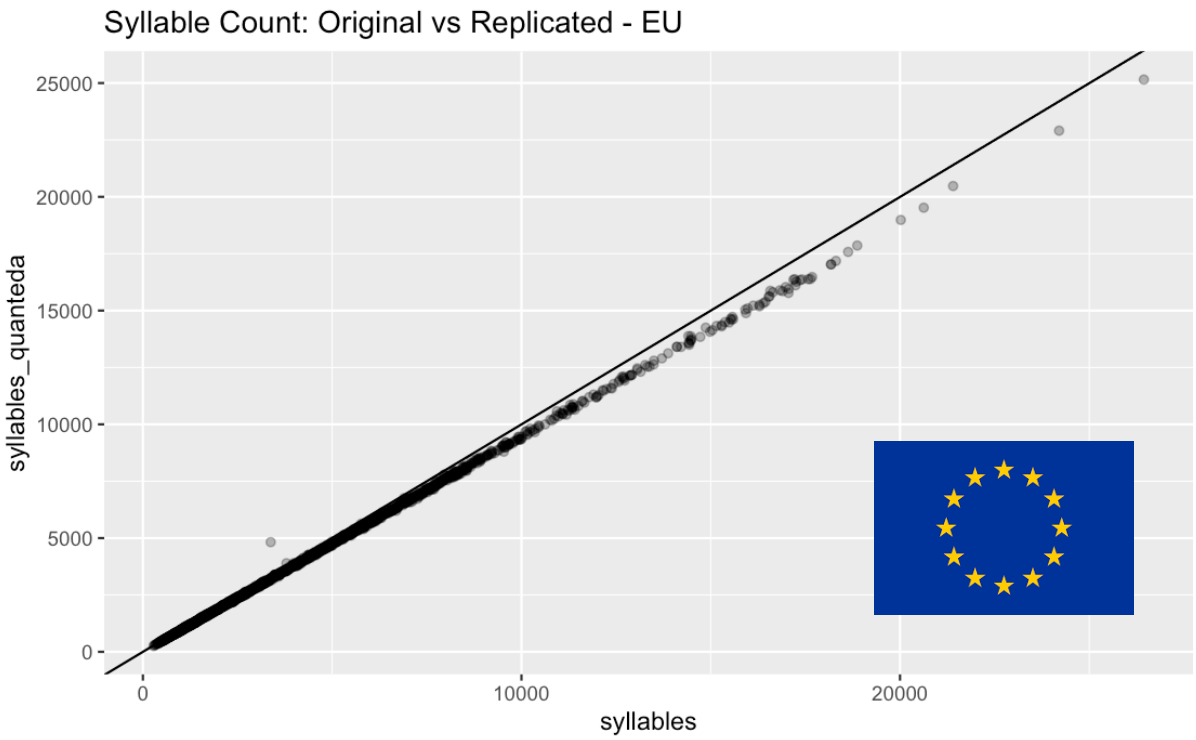


Comparing Sentence Count: Replication vs Authors





Comparing Syllable Counts: Replication vs Authors



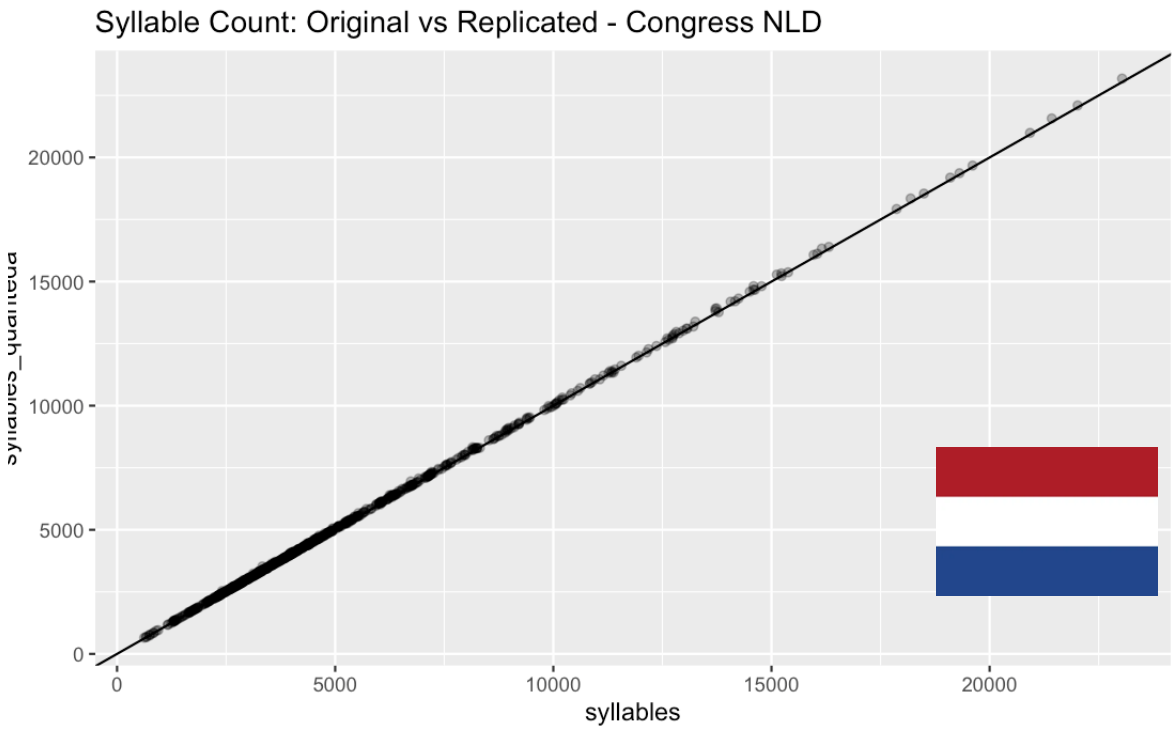
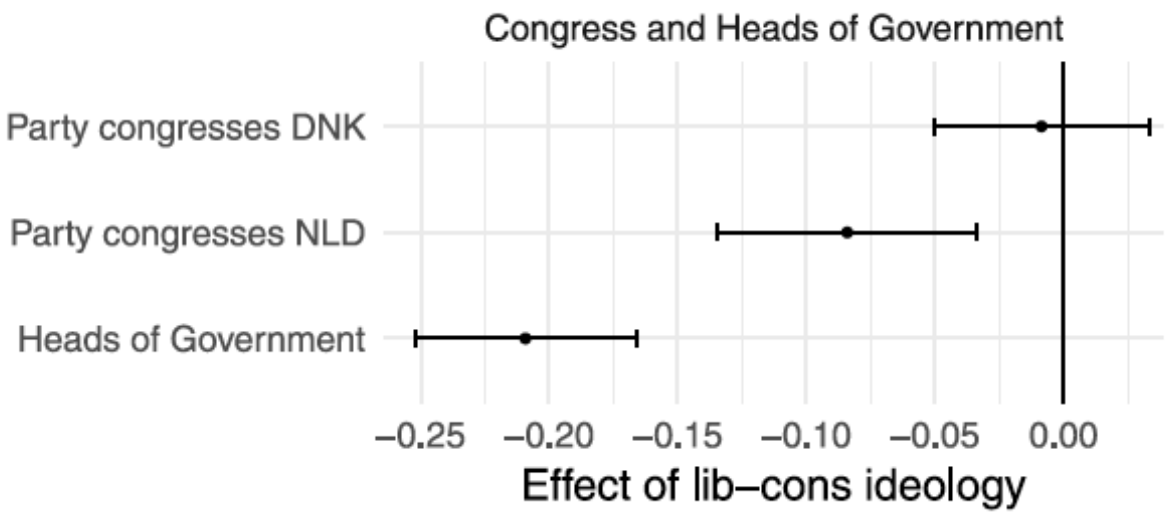


Figure 2a – OLS Regression: Complexity on Ideology (Libs-Cons Social Ideology)



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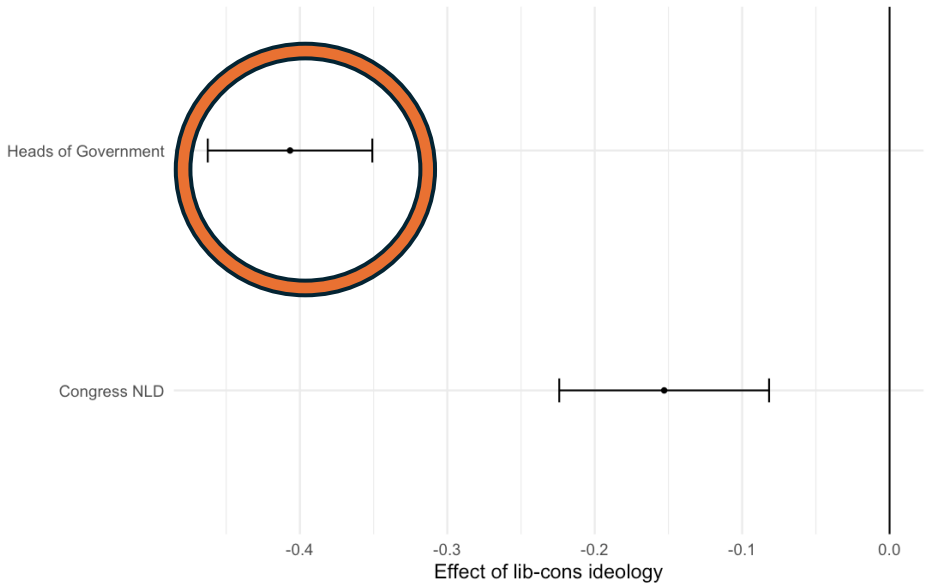
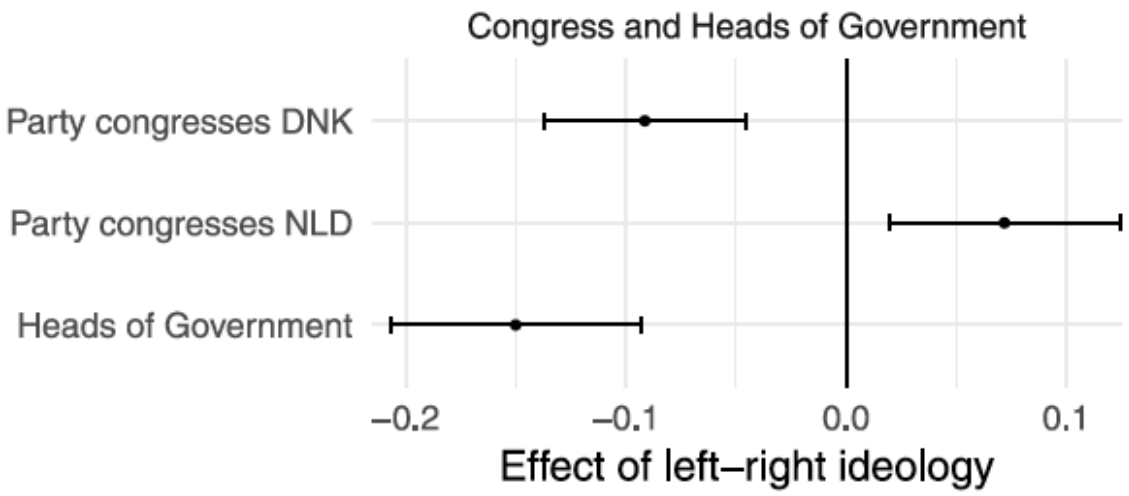


Replication

Figure 2b – OLS Regression: Complexity on Ideology (Left-Right Economic Ideology)



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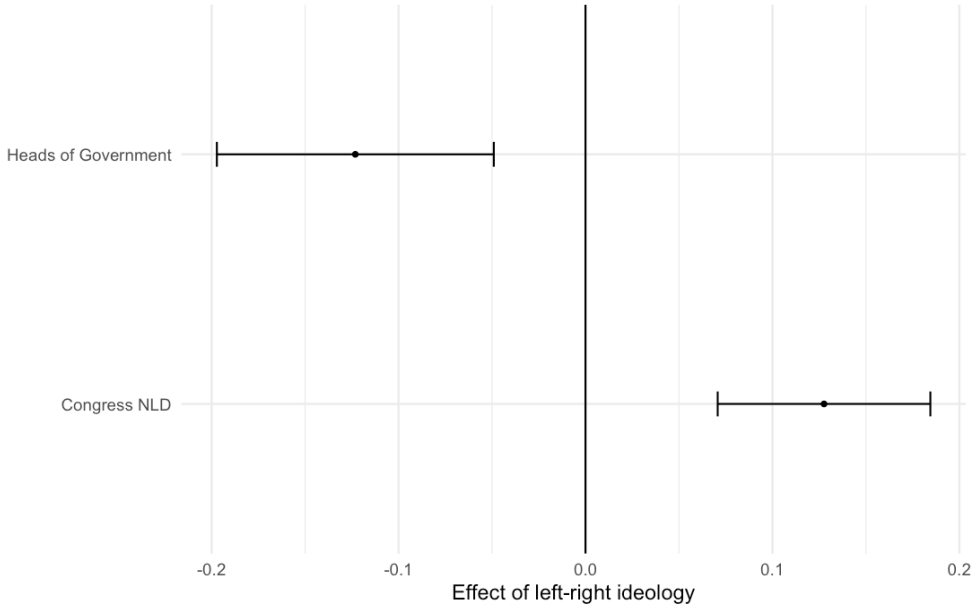
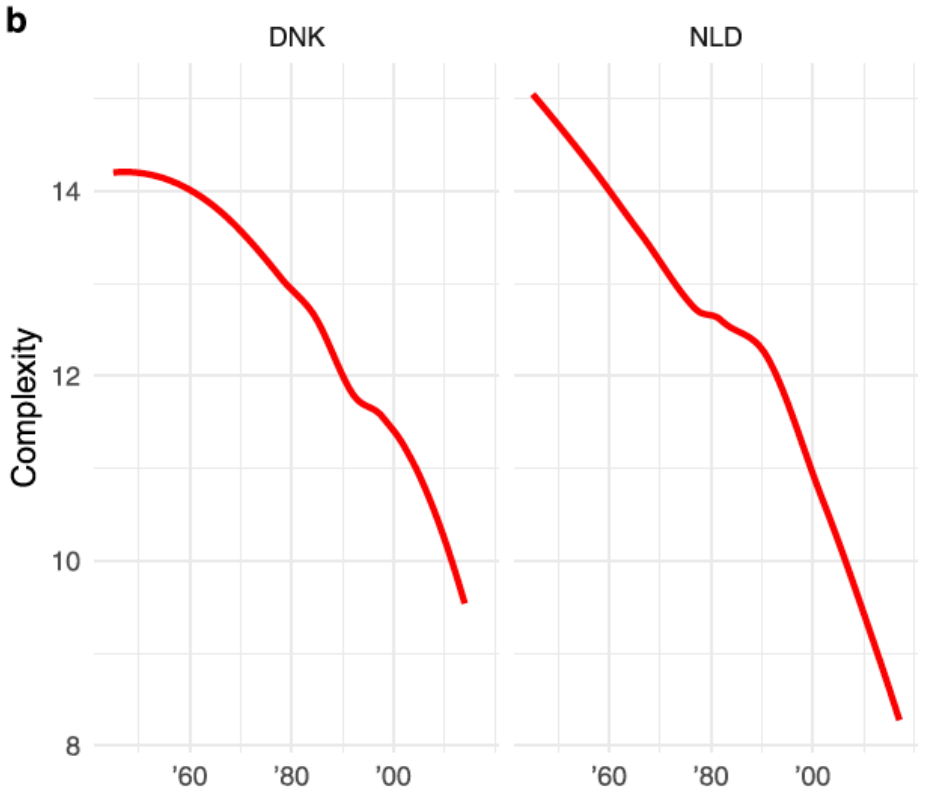


Replication

Figure 3 – OLS Regression: Complexity on Time



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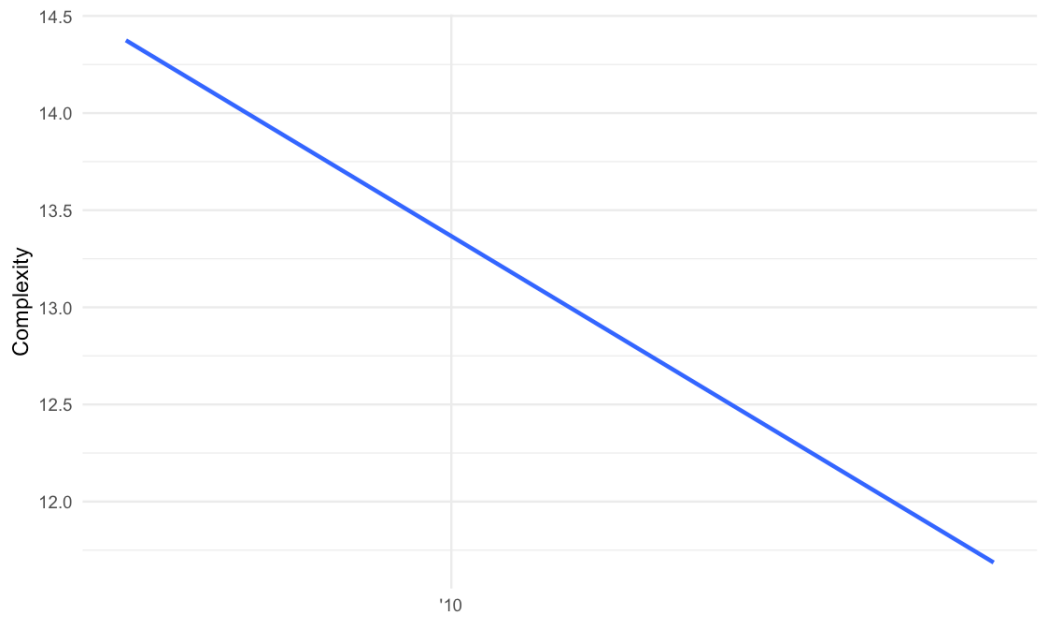
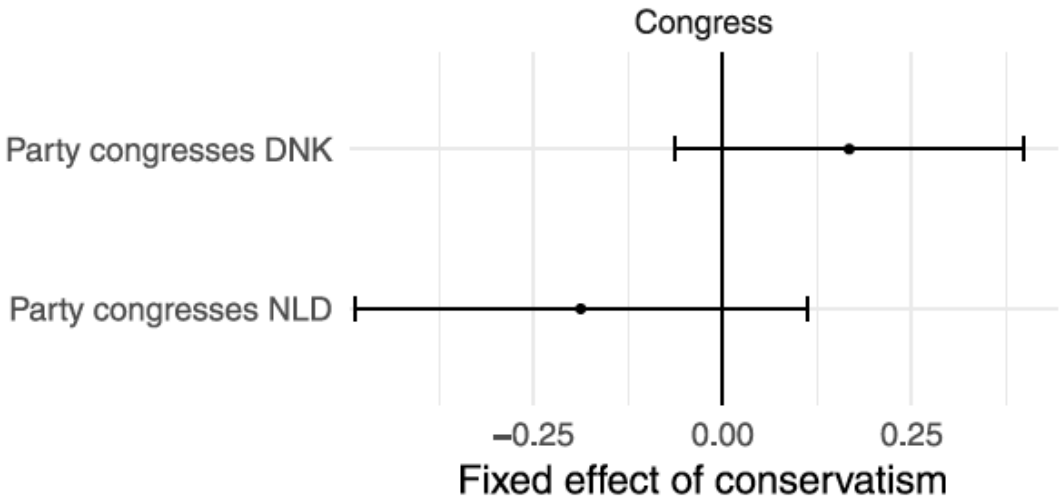


Replication

Figure 4a – OLS Regression: Complexity on Ideology (Libs-Cons) with Fixed Effects



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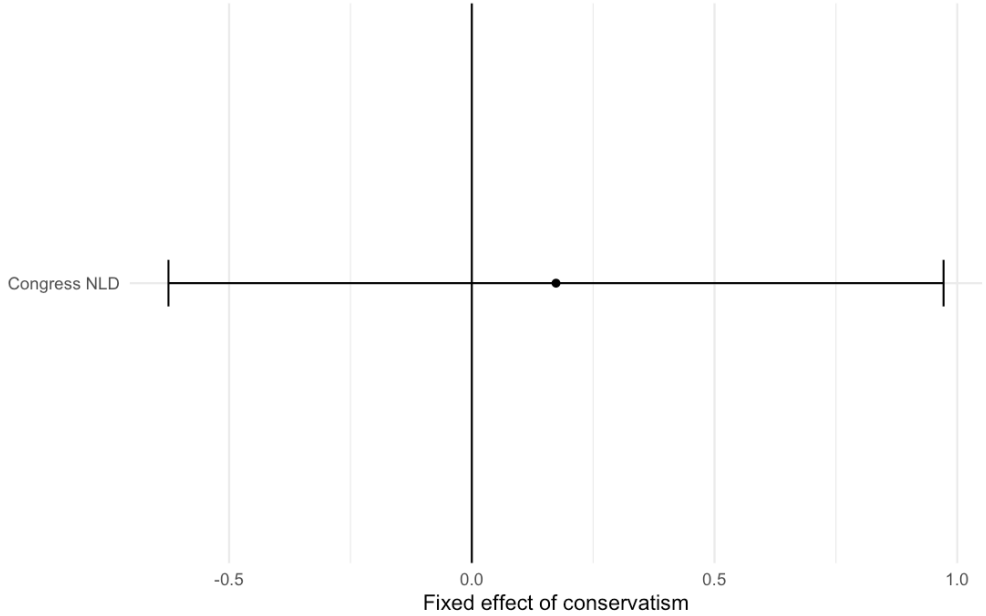
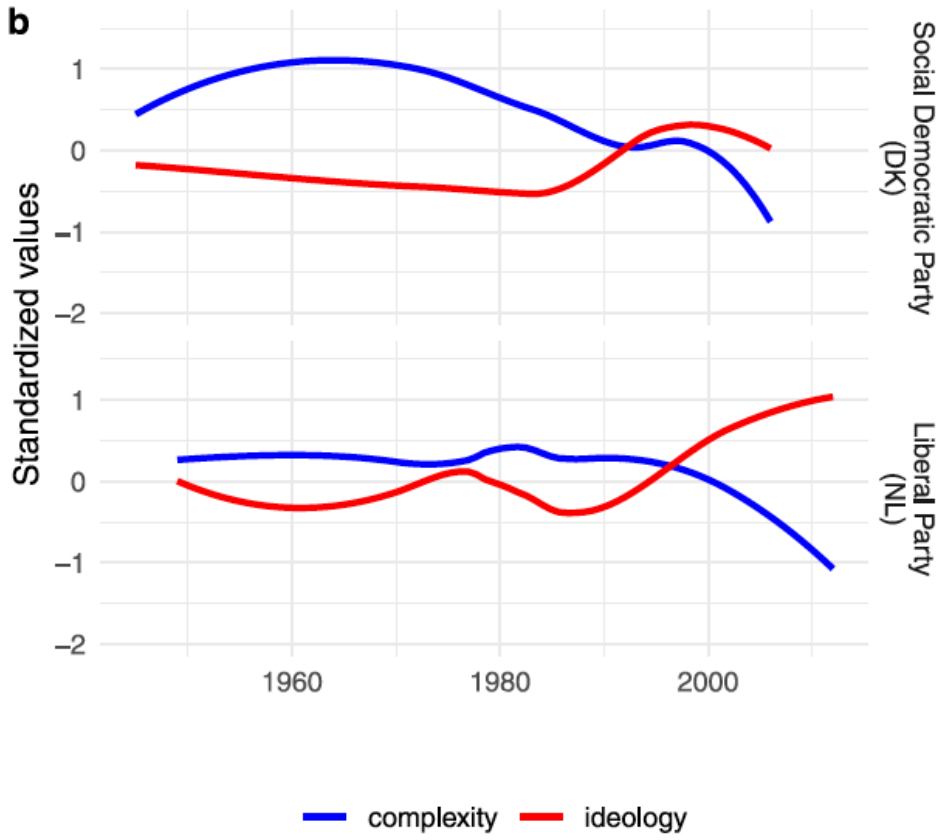


Replication

Figure 4b – Average Complexity and Ideology Over Time (Select Liberal Congress Parties)



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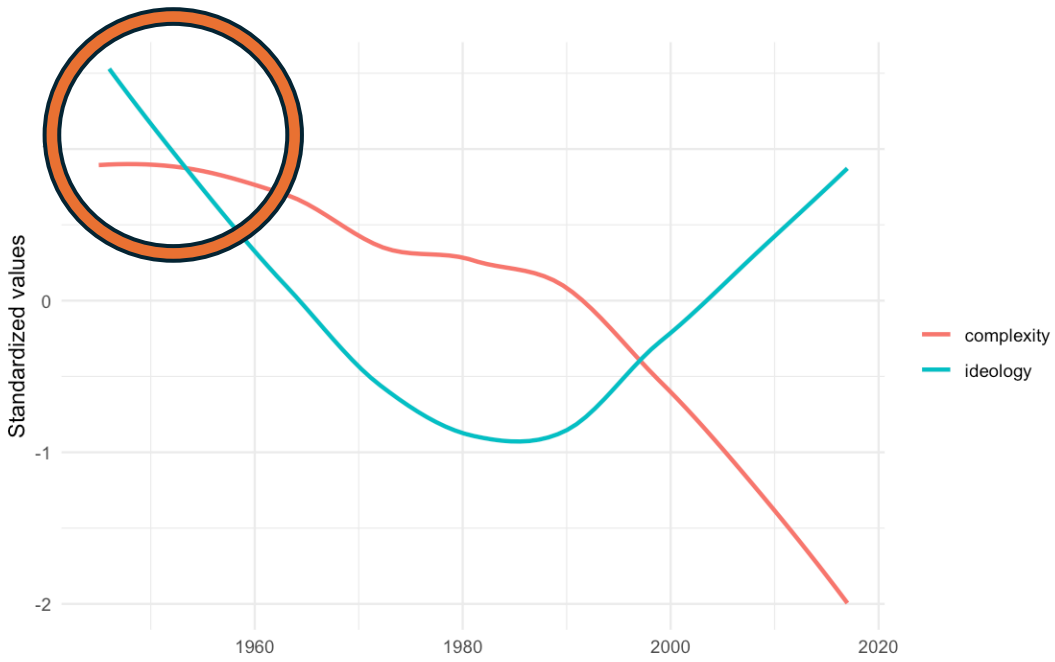

Replication

Replication Autopsy

- **What worked well?**
  - Tables/Figures Code
- **What didn’t work well?**
  - Missing full unprocessed speeches (full corpora, parliament)
  - Flesch Grade scores – how were they created? Syllables and sentences?
  - Zapatero & rajoy – how were speakers selected?
- **Where did I get stuck?**
  - Replicate/validate ideology scores (Manifesto Project)

Replication Extension

- **Change Unit of Analysis**
  - Paragraph level/sentence block
- **Human Validation of Flesch Grade Complexity**
  - Cross cultural research team
- **Add Structural Topic Models**
  - Ideology X Topic → Complexity
  - **1) Topic selection**: Do left vs right speakers emphasize different topics?
  - **2) Linguistic style:** Are those topics expressed with different complexity?
  - **3) Interaction:** Does ideology *change how complex a topic is discussed*?

Overall Assessment

- **Difficulty of replication projects**
- **Computerized scale vs human nuance - validation**
  - Syllables, word counts, sentences
- **Mechanisms matter**